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The Abolition of Immigration Restrictions and the Performance of Firms and Workers: Evidence from Switzerland[†]

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We study a reform that granted European cross-border workers free access to the Swiss labor market and had a stronger effect on regions close to the border. The greater availability of cross-border workers increased foreign employment substantially. Although many cross-border workers were highly educated, wages of highly educated natives increased. The reason is a simultaneous increase in labor demand: the reform increased the size, productivity, and innovation performance of skill-intensive incumbent firms and attracted new firms, creating opportunities for natives to pursue managerial jobs. These effects are mainly driven by firms that reported skill shortages before the reform. (*JEL* J15, J23, J24, J31, J61, K37)

Policies that open the labor market to foreigners are often opposed by natives on the ground that they could harm native labor market opportunities. Yet, free mobility of workers means more opportunities for businesses to hire a wider variety of skills. Firms usually welcome a less restricted access to foreign workers.[1] If firms benefit from open borders through increased productivity and growth, this may counteract

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[1] In a survey by BAK (2013), 75 percent of all employers in Switzerland, the country analyzed in this paper, consider access to foreign workers as "important," "very important," or even "indispensable" for their competitiveness and profits.

the effects of increased labor market competition and expand job opportunities for native workers. However, our knowledge on how immigration policies affect firms' success, and whether such effects shape the labor market effects of immigration, is limited. This study attempts to extend our knowledge on the labor market effects of and firms' responses to opening the border.

We study a far-reaching and controversial policy change: the complete removal of all immigration restrictions for workers from the European Union (EU) in Switzerland when the latter introduced the principle of the "free movement of persons." This principle allows EU citizens to access jobs with no restrictions within the territory of member states. This paper analyzes the consequences of one central aspect of the reform: the removal of all preexisting restrictions on European *cross-border workers* (CBW). CBW are employed in Switzerland, live in its neighboring countries (Italy, Germany, Austria, and France), and commute across the border for work. CBW were already a sizable group in Swiss regions near the border prior to the policy changes. However, there were several administrative hurdles to hiring them. For example, CBW were subject to a bureaucratic process that aimed at ensuring that firms only hire them if they did not find an equally qualified resident worker (the so-called priority requirement). These prior restrictions were gradually abolished after the announcement of the reform in 1999. The priority requirement was abolished in 2004, and CBW thus gained free access to Swiss labor markets close to the border.

How did the greater availability of CBW affect Swiss workers and firms? To study this question, we leverage the fact that the greater availability of CBW had stronger effects on firms and native workers close to the border. In locations further than 30 minutes driving distance from the border, employment of CBW remained negligible. One reason is that CBW could not be employed outside the "border region" until 2007, a clearly defined set of municipalities close to the border. Another reason is that CBW rarely work in places located far away from the border simply because they are commuters from abroad and not residents of Switzerland. Empirically, we thus compare changes in outcomes in labor markets close to the border with changes in outcomes in labor markets further away from it, distinguishing a pre-treatment (before 1999), a transitional (1999–2003), and a free movement phase (2004 onward). This difference-in-differences (DiD) strategy has important advantages compared to many previous studies in the immigration literature because both the increase in the availability of foreign workers and its uneven regional impact are a direct consequence of the exogenous change in the policy related to CBW.[2] Our analyses are based on data from a large-scale employer survey conducted between 1994 and 2010, panel data containing the universe of Swiss establishments from the Business Censuses 1991–2011, and a series of firm-level innovation surveys conducted between 1996 and 2013.

We first show that, between 1999 and 2010, the labor market liberalization for CBW produced a net increase of foreign workers equal to 10 percentage points

[2] Studies on the firm and labor market effects of immigration are typically based on the so-called area approach, and isolate supply-driven variation in immigration into regional labor markets by applying a "shift-share" instrumental variable approach. The approach hinges on the assumption that historical immigrant settlement patterns are uncorrelated to the regional distribution of current unobserved labor demand shocks. This assumption is not always plausible (see Jaeger, Ruist, and Stuhler 2019 for a discussion).

of the total 1998 employment in municipalities within 15 minutes travel time to the border. The increase was most pronounced in the post-2004 period, and two-thirds of all new CBW were highly educated. We also observe a small disproportionate increase in employment of permanent *resident immigrants* close to the border. While Switzerland also removed all barriers for EU immigrants in the process of the reform, this aspect of the reform affected all regions independently of the distance to the border. The finding thus suggests that the greater availability of CBW complemented and crowded in resident immigrants from EU countries.

We then document that the greater availability of CBW did not have a statistically significant negative effect on average employment or wages of Swiss native workers. On the contrary, we find robust evidence that the reform *increased* wages of highly educated native workers by around 5 percent. For this group, we can also rule out negative effects on employment. In fact, based on the comparison with one control group, their employment increased as well. We also find weak evidence for negative effects on less-educated natives, mostly in less skilled services, but these estimates are not robust across specifications and mostly lack the statistical precision to rule out zero effects. As many of the incoming workers were highly educated, these results cannot be rationalized by a simple model with high- and low-skilled labor in which immigration represents a pure supply shift (as in, e.g., Borjas 2003).

Based on a search and matching framework with heterogeneous labor, we argue that a greater availability of CBW can have limited displacement effects because it may make it easier for firms to find certain skilled workers that were hard to find before, thus generating incentives to create new jobs for skilled workers. In such a framework, the positive wage effects on highly educated natives can be rationalized if the policy additionally increased firms' productivity, innovation performance, or capital formation.

We present four pieces of evidence in line with these predictions. First, we show that the inflow of CBW was largest in high-tech manufacturing and the knowledge-intensive business services, i.e., industries that depend on the availability of skilled workers. We find that the positive wage effects on natives are similarly concentrated in these skill-intensive sectors because the reform increased labor productivity of incumbent firms in the sectors. Second, we show that the reform boosted firm expansion by relaxing prior constraints to recruit skilled workers. In particular, we observe substantial gains in labor productivity in incumbent firms that reported to be constrained by a lack of specialized personnel or limited by labor market regulation for foreign workers before the reform. Third, we show that the free movement policy increased R&D employment, patent applications, and product innovations. Again, these effects are concentrated in firms that reported scarcity of R&D workers before the reform. Fourth, we find evidence that the opening of the border led to net entry of establishments, a process that started early during the reform. In sum, these firm effects created opportunities for natives to grow professionally and their likelihood to work in top managerial positions increased. These transitions into high-paying management explain roughly one-third of the positive wage effects for highly educated natives.

We provide a large set of robustness checks that corroborate the causal interpretation of these findings. Most importantly, we show that our results are not driven by firms and industries that were most affected from the trade liberalizations between

Switzerland and the European Union that occurred largely simultaneously with the changes in the commuting policy. We also discuss whether regions close to the border may have partly grown *at the expense* of regions further away. Indeed, some of our firm-level findings are consistent with such an interpretation. However, we do not find evidence for a systematic mobility response of natives that would likely arise if one region benefited at the expense of other regions. Although these results are not ultimate empirical proof, they reduce concerns that our DiD estimates mainly reflect relative rather than absolute causal effects.

Ours is one of the first studies exploiting changes in policies for cross-border commuters to study the effects of immigration.[3] The closest precursor to this paper is Dustmann, Schönberg, and Stuhler (2017), who analyze the labor market effects of the opening of Germany's labor market to Czech cross-border workers in 1991. Dustmann, Schönberg, and Stuhler (2017) show that the inflow of Czech workers had strong negative short-run effects on native employment and smaller but significant negative effects on native wages.

We believe that the different consequences for natives of the opening of the German labor market compared to the Swiss opening are due to differences in the design and economic circumstances of the policy change. First, the Czech inflow was mainly composed of less-educated workers hired in relatively low-skill-intensive industries. In contrast, many new CBW in the Swiss case were highly skilled, and the beneficial effects on natives arose in industries dependent on skilled workers. Second, the policy change in Germany was unexpected, affected regions that had not experienced significant immigrant inflows previously and that had a less developed industrial structure. In contrast, the Swiss policy change was announced early and phased-in gradually, and it had the strongest impact on regions that were used to foreign workers and had a more competitive industry structure. Presumably, the firms were more prepared to match the new workers to jobs in an efficient way. Finally, the German episode took place when both Germany and the Czech Republic underwent a major economic transition. In contrast, Switzerland opened its labor market at a time when unemployment was very low and demand for skilled workers was high. Relaxing the constraints on the supply of skilled workers appears to have benefited firms that had suffered from a lack of skilled labor before the reform.

This paper makes three main contributions to the literature. First, it shows that firms' responses to changes in the availability of skilled labor determine the labor market effects of opening the border. Our labor market findings can only be rationalized when accounting for firms' productivity, capital investment, innovation, and specialization responses. So far, "there is very little tradition for considering firms in analyses of immigration" (Kerr, Kerr, and Lincoln 2015, p. S148). In fact, very few previous papers analyzed firm and labor market effects of immigration jointly.[4] Our study thus contributes to the literature on the impacts of skilled immigration on productivity, innovation, and production technology in the receiving country (see Kerr, Kerr, and Lincoln 2015 for an overview). Studies at the regional level (e.g., Hunt and

[3] The idea to exploit the changes in Switzerland's commuting policies to study the effects of immigration was also pioneered in two policy reports by Henneberger and Ziegler (2011) and Losa, Bigotta, and Oscar (2014).

[4] Examples of empirical papers that study the labor market and firm effects of immigration jointly include Dustmann and Glitz (2015) and Aksu, Erzan, and Kirdar (2018). Waugh (2018) provides a theoretical analysis of how the dynamics of the firm affect economic outcomes from changes in immigration policy.

Gauthier-Loiselle 2010; Peri, Shih, and Sparber 2015a, b) or that focus on inventors (Moser, Voena, and Waldinger 2014) tend to find positive impacts on productivity and innovation. Firm-level studies examining these links are still rare, focus mostly on the United States,[5] and reach conflicting conclusions.[6]

Second, our study directly informs policy makers about the potential economic benefits of the principle of free movement of persons. This is relevant against the background of mounting opposition to free labor mobility in Europe, which culminated in 2016 with Britain's decision to leave the European Union.

Third, our study is one of the first to rigorously evaluate the consequences of a policy that permanently removed all barriers to labor market access for a group of foreign workers. Our variation is thus different from many other quasi-experimental papers that focus on temporary push-driven surges in immigration, where a large number of immigrants are exogenously placed in specific local labor markets. The permanent change in immigration policy analyzed here rather leads to a gradual matching of new CBW to jobs, and plausibly has a direct effect on firms' incentives to create jobs. Unlike regular migrants, however, CBW do not live in the host country, muting possible effects of immigrants on demand for local nontradables, housing, and social services. While changes in cross-border commuter policies thus allow to more cleanly identify the effects of changes in the availability of workers, they only identify a part of the effects of regular migration. Arguably, the fact that consumption-side effects are unlikely in our setting makes the lack of evidence for displacement effects even more striking.

I. The Immigration Reform

The process of opening the Swiss labor market to citizens from the European Union started with the signing of the bilateral agreements between the European Union and Switzerland on June 21, 1999. The so-called Agreement on the Free Movement of Persons (AFMP) introduced free worker mobility among the signing countries. The relevant details of this agreement were publicly announced in Switzerland in late 1998. After the treaty had been signed, it required the approval of the Swiss electorate, which accepted it in a national referendum in May 2000 with an approval rate of 67.2 percent. The European Parliament and each EU member state also approved the treaty in the year 2000. The AFMP was enacted in June 2002, one-and-a-half years later than planned at the time of the first announcement. Given the timing of the reform, anticipatory effects of the reform are possible from 1999 onward.[7] Given the political circumstances, it appears very unlikely that the

[5] The exceptions are Paserman (2013), who examines how immigration from the former Soviet Union affected Israeli manufacturing firms in the 1990s, and Mitaritonna, Orefice, and Peri (2017), who study the impact of the local concentration of immigrants on productivity of French manufacturers.

[6] The US studies generally focus on evaluating the effects of the H-1B program. The results in Ghosh, Mayda, and Ortega (2014) and Kerr and Lincoln (2010) suggest that greater access to H-1B workers generally increases the size, productivity, and innovation performance of firms that rely heavily on H-1B visas. Doran, Gelber, and Isen (2015), on the other hand, find that winning an additional H-1B worker has no effect on patenting and firm size but increases profits and crowds out resident workers. Kerr, Kerr, and Lincoln (2015) find that hiring young skilled immigrants increases firms' skill intensity but their evidence regarding firm size is inconclusive. Similarly, Olney (2013) finds a small impact of (low-skilled) immigration on employment within existing establishments.

[7] The relevant details of the reform were not public knowledge before 1998, and the success of the negotiations was uncertain prior to a breakthrough achieved only in 1998. In fact, even in 1997 and early 1998, several members

local economic conditions of the regions most affected by the agreements were a consideration in the timing and the content of the treaty.[8]

Table 1 provides a timeline for the step-wise introduction of free movement of persons. The table distinguishes three reform phases (the pre-reform, the transition, and the free movement phase) and two types of foreign workers: permanent resident immigrants (or immigrants for short) and CBW. The shading of the table highlights the restrictiveness of the regulations for the respective worker category. The table shows that permanent resident immigrants from EU countries had been subject to yearly national quotas set by the federal government before the reform and to an admission process very similar to the one for CBW detailed below. These restrictions were removed starting in 2002. EU immigrants gained free and full access to the Swiss labor market with the abolition of annual quotas in 2007. Legally, these changes affected all regions in Switzerland equally.

The table also shows when the barriers to hiring CBW were lifted. These changes only affected municipalities in the *border region* (BR) in the years between 1999 and 2004. The reason is that employment of CBW remained restricted to BR until 2007, as it was before the reform. Figure 1 illustrates the geographical split of Switzerland into the BR (in gray) and the rest of Switzerland, the non-border region (NBR, in white). The BR had been defined in bilateral agreements between Switzerland and its neighboring countries signed between 1928 and 1973. The frontier between BR and NBR remained unchanged in the course of the reform, and it does not follow cultural or religious border, nor cantonal or other administrative borders.

The liberalizations for CBW *within* the BR occurred in two steps. In the *transition phase* that started in 1999, cantonal offices, which were responsible for handling applications for CBW, gained more discretion for doing so. Anecdotally, they exploited this to handle CBW applications in a less stringent manner.[9] Several former restrictions were lifted in 2002. First, the recruitment area for CBW was expanded to the entire neighboring countries of Switzerland. Prior to 2002, Swiss firms could only hire CBW who had lived for at least six months in specific municipalities close to the border to Switzerland. Hence, the change effectively allowed workers from the interior of a neighboring country to migrate to the Swiss border to take advantage of the labor market access. Second, new cross-border permits were now generally valid for five years and no longer linked to a specific job. Before 2002, cross-border permits were formally limited to one year and ended with the termination of a work contract, restricting the geographical and occupational mobility of CBW. Third, CBW were only required to commute to their place of residence weekly rather than daily.

of the Swiss parliament expressed their concerns that the negotiations could fail.

[8]One reason is that the federal government, not the cantons, negotiated over the AFMP. Another reason is that introducing the free movement of persons was not championed by the Swiss government but a political concession to the EU. At the beginning of the negotiations in 1993 that led to these agreements, the Swiss government tried to avoid a full-fledged version of free worker mobility. As the EU insisted on full labor mobility, a breakthrough in the negotiations was only reached when both parties agreed that the free labor mobility would be implemented step-wise and included further safety measures.

[9]Conversations with representatives from cantonal migration offices revealed that there was a more relaxed handling of new CBW applications after 1999, and particularly after the national referendum on May 21, 2000, as it was clear that eventually CBW would be the first to gain unrestricted access to the BR.

Table 1—The Different Phases of the Introduction of Free Movement of Workers

			Cross-border workers		Immigrants
Phase	Year	Event	Border region	Non-border region	Both regions
Pre-reform	1995		Admission process (priority requirement), further restrictions	No access	Admission process, annual quotas, further restrictions
	1996				
	1997				
	1998	Announcement			
Transition phase	1999	AFMP signed	Anticipatory effects possible		
	2000	Referendum			
	2001				
	2002	AFMP enacted	Abolition of further restrictions		Higher quotas, further changes[a]
	2003				
Free movement phase in border region	2004	Liberalization in border region	Free		Abolition of admission process
	2005				
	2006				
	2007	Full liberalization		Free	Free
	2008				

[a] Extension of durations of several residency permits. Allowance of family reunion for most permit holders.

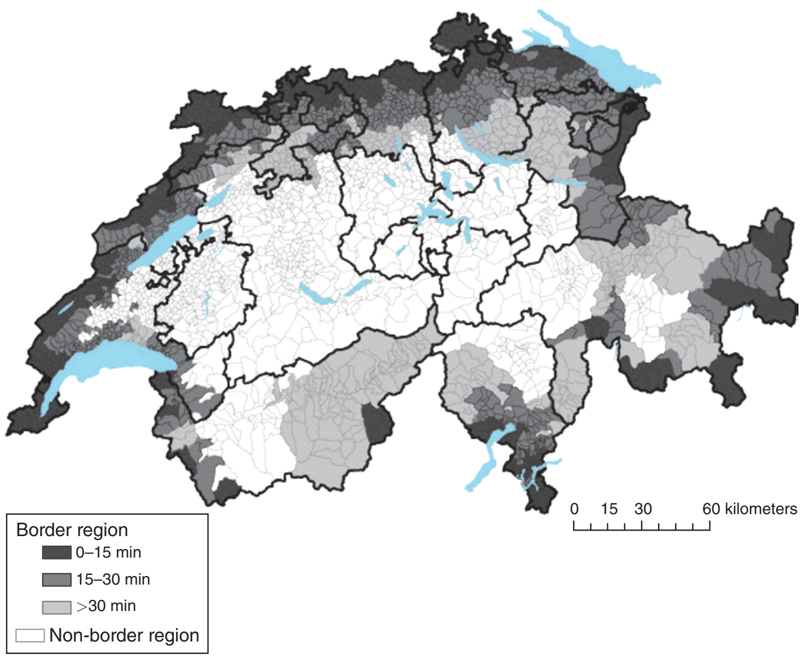

Figure 1. The Border and Non-Border Region and Travel Distance to the Border

Notes: This figure depicts municipalities in the border region in three different shades of gray and those in the non-border region in white. Within the border region, we distinguish three regions according to their travel time by car to the nearest border crossing. The black lines denote cantonal borders.

The *free movement phase* began in mid-2004 when firms in the BR gained full and free access to CBW. Switzerland dropped the bureaucratic admission process for CBW that had been in place before. In particular, Swiss firms had to provide

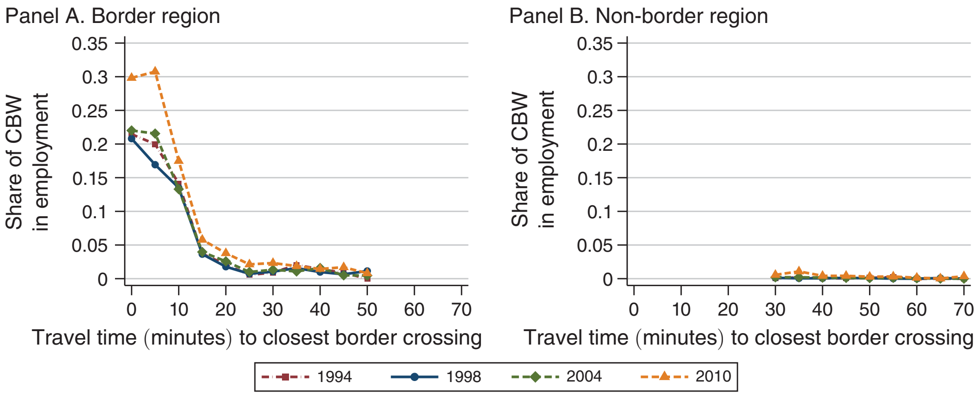

Figure 2. Number of Cross-Border Workers Relative to Total Employment in 1998 by Travel Distance to the Border

Notes: The figure plots the number of cross-border workers relative to total employment in 1994, 1998, 2004, and 2010 separately for the border region (panel A) and the non-border region (panel B). Municipalities are grouped into bins of 5 minutes according to their travel time by car to the next border crossing. Bins with very a small number of total workers are omitted, i.e., those with travel time above 50 minutes in the border region and those between 13 and 30 minutes in the non-border region.

evidence that they had not found, "within an appropriate period of time," resident workers who were willing and capable of filling their vacancies. This regulation, called the "priority requirement," imposed direct recruitment costs for firms hiring CBW by requiring them to go through a relatively lengthy admission process.[10] In June 2004, hiring CBW in the BR became as easy as hiring Swiss workers.

The number of CBW employed in the BR increased substantially in the years of liberalization. Importantly, this increase in CBW was strongly concentrated in labor markets close to the border. Figure 2 uses data from the Swiss Earnings Structure Surveys to plot the share of CBW in total employment separately for the BR and the NBR. Municipalities are grouped into bins of 5 minutes travel-time by car to the nearest border crossing. The figure shows that CBW were almost exclusively employed in municipalities in the BR between 0 and 30 minutes from the border, both before and after the reform. The figure also reveals the *change* in the employment share of CBW over time. The change was very small and sometimes even negative in the pre-liberalization period (i.e., between 1994 and 1998). During the transition period (1999–2003), the share increased slightly, but only in municipalities close to the border. The increase in the share is largest in the free movement phase (i.e., between 2004 and 2010), and it is larger closer to the border. We also show the increase in

[10] When hiring a CBW, firms had to prepare an application detailing the job requirements of their vacancy and the working and contract conditions offered. Moreover, firms had to provide proof that they had searched unsuccessfully for a worker within Switzerland for a certain number of weeks. The application had to be sent to the cantonal and federal migration offices. The processing of the application lasted about one to three months. The migration offices evaluated each application individually, notably by comparing the job requirements with information on the qualifications of residents registered as unemployed. Today, the direct costs for Swiss firms to recruit workers from outside the European Union are estimated to be about 10 to 20 times larger than those for recruiting EU workers (BSS Volkswirtschaftliche Beratung 2013). This is relevant, as hiring non-EU workers is regulated similarly today as hiring CBW before the reform.

the employment share of CBW in the NBR (panel B of Figure 2) between 2004 and 2010, but this increase is quantitatively very small.

We thus focus on firms and municipalities close to the border within the BR. Due to the limited employment of CBW in the NBR, we do not exploit the switch from no to free access for CBW in the NBR in 2007. To allow for transparent empirical analyses, we analyze the reform by partitioning Switzerland into four regions.[11] Based on the evidence in Figure 2, municipalities and firms located 0–15 minutes away from the border are considered as strongly treated; those between 15 and 30 minutes as weakly treated; and firms and municipalities over 30 minutes within the BR and those in the NBR will form the two control groups. Since both control areas are officially treated (one in 2004 and one in 2007), our estimates are, if anything, biased downward if the policy had effects on regions located more than 30 minutes away from the border. In general, we present the main results using either of the two control groups because there are no strong a priori reasons to prefer one control group over the other.

Three features of the AFMP have important implications for our research design and the interpretation of our results. First, the AFMP was part of a package of seven agreements negotiated at the same time. In general, these other bilateral agreements pertained to harmonizations in specialized fields (e.g., air and land traffic, agriculture, or research cooperation) that likely had very limited effects on the outcomes that we study in this paper.[12] However, one agreement reduced non-tariff barriers to trade between Switzerland and the European Union. It is conceivable that these trade liberalizations affected regions close to the border more than the regions further away. A central goal of our robustness checks is to demonstrate that the effects identified with our strategy are not driven by this simultaneous policy change.

Second, the AFMP also lifted all restrictions for Swiss residents to work in neighboring countries. Yet, the change in employment of CBW in Switzerland was about nine times larger than the change of CBW from Switzerland working in neighboring countries. This reflects the much higher nominal wages and cost of living in Switzerland that make it very unattractive to live in Switzerland while working abroad.[13] Our analyses thus abstract from the fact that the reform lifted restrictions on Swiss CBW.

Third, we interpret the reform as *increasing the availability of CBW* in regions close to the border. This is not equivalent to interpreting the reform as an increase in the supply of CBW. The reform also plausibly affected the CBW already working in Switzerland prior to the reform, as these CBW enjoyed increased geographical and occupational mobility. Moreover, like any reform that reduces restrictions

[11] In Beerli and Peri (2018) and Ruffner and Siegenthaler (2017), we show that the results are similar if we use finer intervals and differently defined regions, or if we exploit the continuous nature of the travel distance.

[12] This also holds for the agreement pertaining to research cooperation, which laid the foundation for Switzerland's full participation in the research framework programs of the EU. The programs are targeted at scientific institutions. Moreover, the agreement merely formalized Switzerland's former affiliation within the program, and thus had limited impact on research contributions to Swiss institutions. The impacts of the agreement on private-sector innovation outcomes in our sample period are thus likely very limited.

[13] See online Appendix Table A.2. Data from the Eurostat/OECD purchasing power parities (PPP) program suggest that consumer prices were between 23 percent (France) to 34 percent (Germany) lower in neighboring countries compared with Switzerland in 2009. Eurostat's labor cost survey in 2012 suggests that nominal wage costs per hour are between 33 percent (France) to 46 percent (Italy) lower in neighboring countries.

on labor mobility, the reform plausibly increased the ease with which firms could find skilled workers and reduced regulations to hire foreign workers. Therefore, the reform likely had a direct impact on firms' incentives to create jobs (see Section V). Another implication is that we focus on the *reduced-form effects* of the reform throughout, the extent to which permanently opening the labor market for CBW affected resident workers and firms, and abstain from presenting instrumental variable estimates that scale the reduced-form effects with the effect on the employment share of CBW. Such estimates would entail an interpretation of the coefficients as an impact of labor supply changes.

II. Data and Empirical Strategy

A. *Data*

Our empirical analyses are based on three datasets (online Appendix Table A.1 provides an overview). The main data sources for the labor market analysis are the Swiss Earnings Structure Surveys (SESS) 1994–2012. The Swiss Federal Statistical Office has conducted these surveys every two years since 1994. They are a stratified random sample of private and public firms with at least three full-time equivalent (FTE) workers from the manufacturing and service sectors, covering between 16.6 percent (1996) and 50 percent (2010) of total employment in Switzerland. The data include detailed information about workers, their wages and full-time equivalents, their demographic characteristics, and their place of work. We focus on individuals aged 18–65, working in the private sector, with non-missing information for nationality, place of work, education, wages, full-time equivalents, and some other basic demographics.[14] Based on information about workers' residency permits, we distinguish between *native workers*, i.e., those with Swiss nationality; foreign-born workers with a residency permit, which we call *resident immigrants I*; and CBW. Using the SESS, we analyze the reform effects on the number of cross-border and foreign-born workers as a share of total employment, and the effects on full-time equivalents and real hourly wages of natives. We define workers with tertiary education as being *highly educated*. Workers with completed secondary education (such an apprenticeship) and those with primary education are combined in a group of *less-educated*.[15]

Our second data source consists of six waves of the Swiss Business Censuses conducted in 1991, 1995, 1998, 2001, 2005, and 2008 by the Swiss Federal Statistical Office, and the 2011 wave of the register-based Business Census (Swiss Federal Statistical Office 2011). The Business Censuses (BC) constitute a panel dataset covering the *universe* of private and public establishments in Switzerland. Approximately 4 million employed persons in 389,000 workplaces are included in the census of 2008. The data provide us with information on the size (FTE

[14] Online Appendix Section B.1 contains a detailed discussion of the sample construction for the labor market analysis.

[15] There are some a priori reasons to show separate results for these two subgroups. This is the approach followed in Beerli and Peri (2018). For brevity and because of the similarity of the labor market results, we decided to pool the two subgroups in this version. Indeed, previous research suggests that these two subgroups are closely substitutable (Gerfin and Kaiser 2010, Müller and Graf 2015).

employment) and the exact geographical location (geographic coordinates) of all establishments in Switzerland. Until 2008, the censuses were based on mandatory surveys in October. The census 2011 was constructed from administrative data and refers to December.

The third data source is the innovation surveys (IS) of KOF Swiss Economic Institute. These surveys were conducted among Swiss companies between 1996 and 2013 in seven waves. All surveys are based on a representative sample of private-sector firms with at least five FTE employees. The surveys are stratified with respect to firm size and two-digit industry. The IS provide very detailed information on the characteristics of the surveyed firms and a rich set of outcomes such as value added and the number of patent applications filed. However, the data cover only a relatively small sample of firms per wave, and have some of the limitations of voluntary surveys such as reporting errors, attrition, and non-response. The average response rate across all surveys is 35 percent. In addition, the unit of observation is the firm, not the establishment. We thus have to assign multi-establishment firms to the location of the headquarter.[16] As a result, our estimates with the IS are not very precise.

B. *Descriptive Statistics*

Panel A of Table 2 compares the pre-reform characteristics of workers and firms of the four regions that we compare in our DiD estimations: highly treated regions and weakly treated regions (within 15 and 15–30 minutes to the border in the BR, respectively), and the two control regions (more than 30 minutes to the border within the BR and the NBR). The table suggests that the four groups are quite comparable in terms of labor market size, important worker characteristics, and workers' mean log hourly wages. While the employment of CBW was much larger in the treated regions before the reform, the employment share of resident immigrants was similar. The panel also suggests that neither of the two control groups is clearly more comparable to the highly treated region. Similar comments apply if we compare the characteristics of establishments and firms across regions (panels B and C of Table 2). However, we also observe some important differences such as the fact that highly treated establishments are somewhat larger and more likely to be exporters than establishments in the two control groups.

In the online Appendix, we also describe the characteristics of the CBW working in the BR after 1998. Three features are noteworthy (see Table A.3). First, while CBW were on average less-educated than natives before the reform, we observe a large increase in the share of highly educated CBW in the 1998–2010 period (+12.5 percent). Consistent with such high education levels, the employment of CBW grew most in occupations with high and intermediate wage levels.[17] Second, the increase in CBW was largest in IT, R&D, business services, real estate, and, to a lesser extent, in the health sector, suggesting that many new CBW were professionals in science and technology. Third, using a Mincer regression, we find

[16] Online Appendix Section B.2 provides detailed discussions on how we constructed our estimation sample for the two datasets and how we assign firms to BR and NBR.

[17] Beerli, Indergand, and Kunz (2017) show that the increase in tertiary education among new immigrants in Switzerland between 1990–2010 was a response to long-term, technology-driven increase in the demand for skills.

Table 2—Firm and Worker Characteristics Prior to the Reform by Region

	Border region						Non-border region	
Travel time to border:	≤15 min		15–30 min		>30 min			
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Panel A. Demographic characteristics (SESS)								
Share highly educated	0.18	(0.38)	0.19	(0.39)	0.17	(0.37)	0.15	(0.35)
Share less-educated	0.82	(0.38)	0.81	(0.39)	0.83	(0.37)	0.85	(0.35)
Age	39.62	(10.92)	39.38	(11.30)	39.13	(11.26)	38.73	(11.41)
Share male	0.61	(0.49)	0.63	(0.48)	0.61	(0.49)	0.60	(0.49)
Tenure (in years)	8.88	(8.56)	8.37	(8.57)	8.49	(8.61)	8.64	(8.60)
log hourly wage	3.50	(0.38)	3.55	(0.37)	3.48	(0.35)	3.45	(0.35)
Share cross-border workers	0.18	(0.38)	0.02	(0.13)	0.01	(0.10)	0.00	(0.03)
Share resident immigrants	0.06	(0.23)	0.05	(0.22)	0.07	(0.25)	0.06	(0.23)
Number of workers	501,496		672,749		259,939		521,943	
Panel B. Establishments (BC)								
Travel minutes to border	7.05	(3.54)	23.36	(4.07)	39.95	(10.79)	54.06	(13.98)
FTE employment	17.84	(67.57)	18.13	(60.60)	15.16	(51.33)	14.76	(47.58)
Share exporter (1995)	0.20	(0.40)	0.18	(0.38)	0.15	(0.36)	0.13	(0.33)
Share importer (1995)	0.30	(0.46)	0.29	(0.45)	0.24	(0.43)	0.23	(0.42)
Share high-tech manufacturers	0.05	(0.21)	0.04	(0.20)	0.05	(0.21)	0.04	(0.20)
Share low-tech manufacturers	0.12	(0.32)	0.11	(0.32)	0.12	(0.32)	0.13	(0.34)
Share in knowl.-intensive services	0.24	(0.43)	0.25	(0.44)	0.23	(0.42)	0.21	(0.41)
Share in not-knowl.-intensive services	0.47	(0.50)	0.46	(0.50)	0.46	(0.50)	0.47	(0.50)
Observations	17,234		22,996		11,086		23,646	
Panel C. Firms (IS)								
Firm age (in years)	45.28	(35.30)	45.44	(37.88)	46.08	(36.21)	51.37	(48.62)
Share of firms with R&D expenditures	0.43	(0.50)	0.51	(0.50)	0.49	(0.50)	0.42	(0.49)
Export share in sales	0.22	(0.33)	0.21	(0.32)	0.20	(0.31)	0.17	(0.30)
Share academics in workforce	0.18	(0.22)	0.19	(0.21)	0.16	(0.18)	0.15	(0.17)
log total sales	16.15	(1.84)	16.50	(1.87)	16.13	(1.84)	16.11	(1.77)
log wage per FTE worker	11.12	(0.52)	11.21	(0.53)	11.13	(0.48)	11.13	(0.47)
Value added per FTE worker (ln)	11.69	(0.56)	11.82	(0.63)	11.74	(0.52)	11.73	(0.55)
Share with high skill shortage	0.17	(0.37)	0.18	(0.38)	0.19	(0.39)	0.17	(0.38)
Share with high R&D shortage	0.17	(0.38)	0.17	(0.37)	0.17	(0.37)	0.15	(0.36)
Observations	932		1,428		610		1,117	

Notes: The table shows descriptive statistics in the border and non-border region. The border region is split into three groups depending on the travel duration to the nearest border crossing. Panel A shows average worker characteristics for the sample of all workers aged 18–64 employed in the private sector from the Swiss Earnings Structure Survey (SESS). Panel B shows establishments characteristics from the Business Census (BC) in 1998 (or 1995, if indicated). Panel C shows average firm characteristics using data from the KOF innovation surveys 1996 and 1999, focusing on characteristics unavailable in the BC. In this panel, entries represent averages per region of all firm-year observations in the two surveys.

that wages of CBW are similar to those of natives after controlling for observable characteristics, suggesting that CBW have comparable labor market skills as observationally similar natives.

C. *Empirical Specification and Identification*

Our basic empirical specification estimates the effects of the greater availability of CBW on Swiss firms and workers by exploiting that the reform interacted with the proximity to the border. We differentiate the effects of the *transition* and the *free movement* phase of the reform by defining the dummies $Transition_t$ and $Free_t$ that are equal to 1 in the years $1999 \leq t < 2004$ and $t \geq 2004$, respectively, and 0 otherwise. We interact these reform variables with two indicators, $I(d_i \leq 15)$ and $I(15 < d_i \leq 30)$, which are equal to 1 if unit i is located within 15 minutes or at 15 to 30 minutes travel time d_i to the nearest border crossing.[18] Using these variables, we estimate the following DiD model for an outcome $y_{i,t}$ of unit i (a municipality, an establishment, or a firm) in year t:

$$(1)\quad y_{i,t} = \beta^T_{d1}\big[Transition_t \times I(d_i \leq 15)\big] + \beta^T_{d2}\big[Transition_t \times I(15 < d_i \leq 30)\big] + \beta^F_{d1}\big[Free_t \times I(d_i \leq 15)\big] + \beta^F_{d2}\big[Free_t \times I(15 < d_i \leq 30)\big] + \alpha_i + \alpha_t + \gamma Controls_{i,t} + \epsilon_{i,t}.$$

In this model, β^{phase}_{d1} and β^{phase}_{d2} capture the impact of a specific phase of the reform on highly and slightly treated units, respectively, i.e., the differential evolution in the outcome $y_{i,t}$ in these groups during the transition phase and free movement phase relative to the control group. We use two different control groups: units in the BR located more than 30 minutes from the border and units in the NBR.[19] The term α_t represents year fixed effects, which absorb the time variation common to all units such as common changes in aggregate prices and demand, as well as the dummies $Transition_t$ and $Free_t$. Finally, α_i represent unit fixed effects that control for preexisting differences between units and regions. Such differences could have been a direct consequence of the established cross-border policy that restricted the hiring of CBW to the BR.

The central identifying assumption of our approach is that we would have observed the same average change in outcomes within units, conditional on controls, in the regions absent the reform. We will test the plausibility of this assumption in several ways. Most importantly, we check how the outcomes in the three relevant regions evolved *before* the reform. To this end, we generalize equation (1) to an event study model. In the case of the data from the SESS, the model takes the following form:

$$(2)\quad y_{i,t} = \alpha_i + \alpha_t + \sum_{t=1994}^{2010} \gamma_{d1,t} I(year = t) \times \big[I(d_i \leq 15)\big] + \sum_{t=1994}^{2010} \gamma_{d2,t} I(year = t) \times \big[I(15 < d_i \leq 30)\big] + \delta Controls_{i,t} + \epsilon_{i,t}.$$

The estimates of the coefficients $\gamma_{d1,t}$ for $t \geq 1999$ reveal the reform effects on the highly treated units; $\gamma_{d2,t}$ reveal the effects on lightly treated units. As the impact of the policy should be 0 before its announcement, we should find that $\gamma_{d1t} = \gamma_{d2,t} = 0$,

[18] The travel distance to the border is computed using information on the location of establishments (BC) and firms (IS). The term d_i is time-invariant because a time varying measure could be endogenous (e.g., represent a relocation of firms in response to the commuting scheme). In the firm regressions, we assign firms to their location in 1998 throughout the estimation period. For the municipality-level specifications, we use the BC 1995 and 1998 to compute the employment-weighted average travel time to the border of the establishments in a municipality. See online Appendix Section B.2 for further details.

[19] If we use the NBR as the control group, we exclude the very few establishments and municipalities located in the NBR within less than 30 minutes to the border. The reason is that these units may be affected by the switch from no to free access to CBW that occurred in 2007 (see online Appendix Table A.1).

for $t < 1998$. We standardize the effects to 0 in 1998 by dropping the indicator for that year from the regression.

Arguably, the main threat to a causal interpretation of our estimates is unobserved factors that are correlated with the timing of the reform and that affect regions differently depending on the distance to the border. Candidate confounding factors are simultaneous reforms (e.g., due to changes in cantonal policies) and unobserved region-specific shocks to prices, demand, or productivity. We partially account for such factors by including linear time trends for each NUTS-II region in our vector of control variables, $Controls_{i,t}$. In Sections IVB and VIE, we also provide several checks that suggest that our results are not confounded by unobserved region- or industry-specific shocks and by the simultaneous changes in trade policies.

Another remark concerns inference. We cluster standard errors at the level of commuting zones, both in the municipality- and firm-level regressions. We thus allow for arbitrary dependence, cross-sectional and over time, between units within the same commuting zone. In online Appendix Tables A.8 and A.17, we compare the standard errors based on this strategy with standard errors clustered at alternative levels and with standard errors based on the spatial heteroskedasticity and autocorrelation consistent (SHAC) variance estimator proposed by Conley (1999), also used by Dustmann, Schönberg, and Stuhler (2017). These alternative standard errors are often substantially smaller than our preferred ones. Our inference is thus conservative.

III. Effects of the Policy on Immigration

Our empirical strategy depends on the idea that the immigration reform affected regions close to the border more. Figure 2 provides descriptive evidence that supports this idea. Using the SESS from 1994 to 2010, we now analyze the exact dynamics of the change in immigrant exposure in municipalities located close to the Swiss border compared to those farther away. Panel A of Figure 3 plots the coefficients $\gamma_{d1,t}$ and $\gamma_{d2,t}$ and their 95 percent confidence intervals in a regression as specified in equation (2) when either the BR 30+ or the NBR constitutes the control group. The dependent variable is the number of total immigrant workers (CBW plus resident immigrants) in municipality i and year t standardized by total employment in 1998, $(CBW_{i,t} + I_{i,t})/Emp_{i,1998}$.[20]

Panel A reveals several important features in the evolution of immigrant exposure by region. First, in the pre-1999 period, none of the estimates is statistically significantly different from zero at the 95 percent significance level. There are thus no differences in the trends of the number of immigrants between municipalities close to the border and either control group before the reform. Second, there is a mild upward trend in the number of immigrants as a share of 1998 employment between 2000 and 2002 in highly treated municipalities. The estimated increase is between 1.5 and 3 percentage points in 2002, depending on the control group, suggesting a small reform effect on immigrant employment in the highly treated regions during

[20] We standardize the number of immigrants by total local employment (native and foreign workers) in the last year prior to the reform, 1998, rather than contemporaneous total employment, as immigration may affect contemporaneous total employment (Dustmann, Schönberg, and Stuhler 2017).

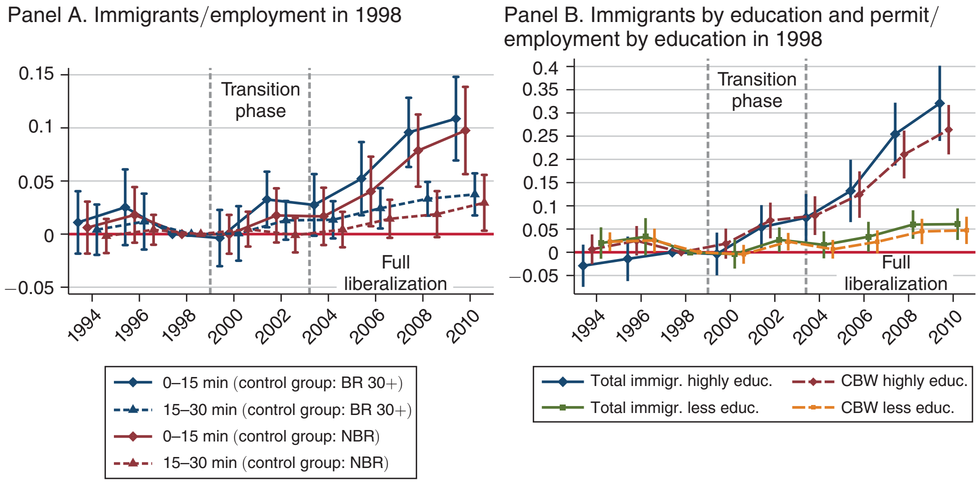

Figure 3. Effects of the Free Movement Policy on the Number of CBW and Resident Immigrants

Notes: The figure shows the effect of the free movement policy on immigration. It plots the estimated reform effects and associated 95 percent confidence intervals for highly treated municipalities (panel A: and weakly treated municipalities) based on the event study model (equation (2)). In Panel A, the dependent variable is the number of immigrants relative to total employment in 1998. We present separate estimates for the two control groups (BR 30+ and NBR). In panel B, the dependent variable is the number of high- or less-educated immigrants (or CBW only) relative to the total number of workers in the same group in 1998. In this panel, the control group is the BR 30+. The regressions are weighted using the total workforce in 1998 in a cell. All regressions account for municipality and period fixed effects, and linear trends per NUTS-II region. Standard errors are clustered on the level of commuting zones.

the transition phase. Third and most importantly, the number of immigrants as a share of 1998 employment grows consistently in highly treated municipalities after 2004, and to a lesser extent, in weakly treated municipalities. By 2010, the reform led to an increase in the share by about 10 percentage points in the highly treated regions. As expected, the estimates indicate that the reform effect was smaller, about one-third, in the weakly treated regions. Fourth, the estimates based on the NBR control group are slightly lower but not statistically different from those based on the BR 30+ control. If we compare the average outcome in the free movement period with the entire pre-reform period, as we do in our baseline DiD model (equation (1)), we estimate an increase in the number of immigrants as a share of 1998 employment by 5.6 (using the BR 30+ as control) and 3.7 (using the NBR) percentage points, respectively (see columns 1 and 2 of Table 3).

The regressions in columns 3–6 of Table 3 show how different immigrant subgroups contributed to this aggregate inflow of immigrants. Columns 3 and 4 of Table 3 show that two-thirds (3.8 of 5.6 percentage points) of the total increase in immigrants in the free movement phase in highly treated regions can be attributed to inflows of CBW. The rest of the excess increase is attributable to resident immigrants, $I_{i,t}$. Considering the legal circumstances and the fact that the pre-reform share of resident immigrants was similar in the relevant regions (see panel A of Table 2), we would expect that the liberalizations for resident immigrants affected treated and control regions similarly. We thus interpret these findings as evidence that the

Table 3—Effects of the Free Movement Policy on the Number of Immigrants by Permit Type and Educational Attainment

	Immigrants by permit type				Total immigrants by education group	
	All		CBW	Residents	High	Less
	(1)	(2)	(3)	(4)	(5)	(6)
$Transition_t \times I(d_i \leq 15)$	0.001	−0.005	−0.001	0.002	0.007	−0.006
	(0.008)	(0.009)	(0.008)	(0.007)	(0.003)	(0.007)
$Transition_t \times I(15 < d_i \leq 30)$	0.002	−0.003	0.002	−0.001	0.004	−0.002
	(0.008)	(0.008)	(0.004)	(0.007)	(0.002)	(0.007)
$Free_t \times I(d_i \leq 15)$	0.056	0.037	0.038	0.018	0.038	0.018
	(0.014)	(0.014)	(0.013)	(0.008)	(0.007)	(0.009)
$Free_t \times I(15 < d_i \leq 30)$	0.022	0.007	0.011	0.012	0.011	0.012
	(0.008)	(0.010)	(0.005)	(0.007)	(0.004)	(0.006)
R^2	0.517	0.516	0.545	0.332	0.548	0.464
Observations	9,585	12,051	9,585	9,585	9,585	9,585
Number of clusters	72	95	72	72	72	72
Control group: BR 30+	✓		✓	✓	✓	✓
Control group: NBR		✓				
Year and area fixed effects	✓	✓	✓	✓	✓	✓
NUTS-II trends	✓	✓	✓	✓	✓	✓

Notes: The table shows the effect of the free movement policy on the number of immigrants based on equation (1). In columns 1 and 2, the dependent variable is the number of immigrants standardized with total employment in 1998. In columns 3 and 4, the dependent variable is the number of CBW or resident immigrants, respectively, standardized with total employment in 1998. In column 5 and 6, the dependent variable is the total number highly or less-educated immigrants, respectively, standardized by total employment in 1998. $Transition_t$ is 1 for the period between 2000 and 2003, whereas $Free_t$ is 1 from year 2004 onward. $(d_i \leq x)$ and $(y < d_i \leq z)$ indicate whether a municipality is located less than x travel minutes or between y and z travel minutes from the next border crossing, respectively. The regressions are weighted using the total workforce in 1998 in a cell. Robust standard errors, clustered by commuting zone, are shown in parentheses.

greater availability of CBW attracted (*crowded in*) resident immigrants, an effect that is statistically significant, as shown in column 4 of Table 3. This interpretation is consistent with the timing of the effect on resident immigrants, which follows the surge in CBW.[21]

Columns 5 and 6 of Table 3 show that most of the increase in CBW is attributable to inflows of highly educated CBW (3.8 of 5.6 percentage points). Indeed, panel B of Figure 3 shows that the inflow of highly educated immigrants was very substantial relative to the preexisting pool of highly educated native workers in 1998: by 2010, the policy increased the number of highly educated immigrants as a share of total employment of highly educated workers in 1998 by roughly 30 percentage points in the highly treated region. In contrast, the growth of immigrants within the group of less-educated workers was much lower (6 percentage points). Complementing this picture, online Appendix Table A.4 shows that the largest contribution to the overall growth in the number of immigrants came from occupations with high pay, such as R&D workers, IT specialists, analysts, and consultants, and, to a lesser extent, from those with middle pay.

[21] The excess increase in the number of resident immigrants close to the border, illustrated in panel B of Figure 3 by the divergence in the effects on total immigrants and CBW, starts in 2006 and thus 2–4 years after the start of the increase in the number of CBW.

In sum, the free movement policy increased the supply of CBW and of other resident immigrants by about 10 percentage points by 2010 in municipalities within 0–15 minutes from the border. Two-thirds of the new CBW were highly educated. As expected, the increase was smaller 15–30 minutes from the border and was most pronounced after 2004, when the labor market in the BR was fully opened to CBW.

IV. Labor Market Effects

A. *Main Results*

In this section, we investigate whether the greater availability of (mainly highly educated) CBW depressed wages or employment opportunities of (highly educated) natives. We analyze wage and employment outcomes of natives jointly as they represent different margins of adjustment, potentially heterogeneous across groups due to group-specific labor supply elasticities or wage rigidities (see Dustmann, Schönberg, and Stuhler 2016 for a discussion).

Part I of Figure 4 provides evidence on the wage and employment effects of the free movement policy on natives. It plots the event study estimates (equation (2)) for the highly treated regions (0–15 minutes) using average log real hourly native wages (panel A) and log total native workers (panel B) as dependent variables, respectively. We present the results using both control groups separately. Online Appendix Table A.5 presents the corresponding point estimates from our baseline DiD model (equation (1)).

Panels A and B of Figure 4 show that natives' wages and employment evolved similarly in the treatment group and in both control groups prior to 1999. This remains true in the reform period: the estimated reform effects are never significant for both outcomes. In terms of employment of native workers, the point estimates are close to 0 based on the BR 30+ control group and slightly negative but non-significant based on the NBR control group. Hence, we neither find statistically significant evidence of a negative effect on average native wages nor a clear effect on native employment despite the substantial increase in immigrant employment in highly treated regions. Our establishment-level regressions presented below strengthen this view: we find no evidence that the substantial increase in employment of foreigners reduced FTE employment of Swiss nationals *within* establishments (see Figure 5).

In panels C and D of Figure 4, we look at the impacts on high- and less-educated native workers separately. The estimates represent the "total" effect of immigrants on wages and employment of each education group of natives. They capture the impact on natives both from the competition with immigrants with similar skills and from complementarity to those with different skills.[22] As the free movement policy produced a larger inflow of CBW with tertiary education relative to those with lower qualifications, the canonical "partial effects" model would imply downward pressure on wages and employment of highly educated natives and possibly positive effects on less-educated natives through complementarity.

[22] See Ottaviano and Peri (2012) for a more formal argument about the estimation of a total effect of immigrants aggregating all the direct competition and indirect complementarity effects from different skill groups.

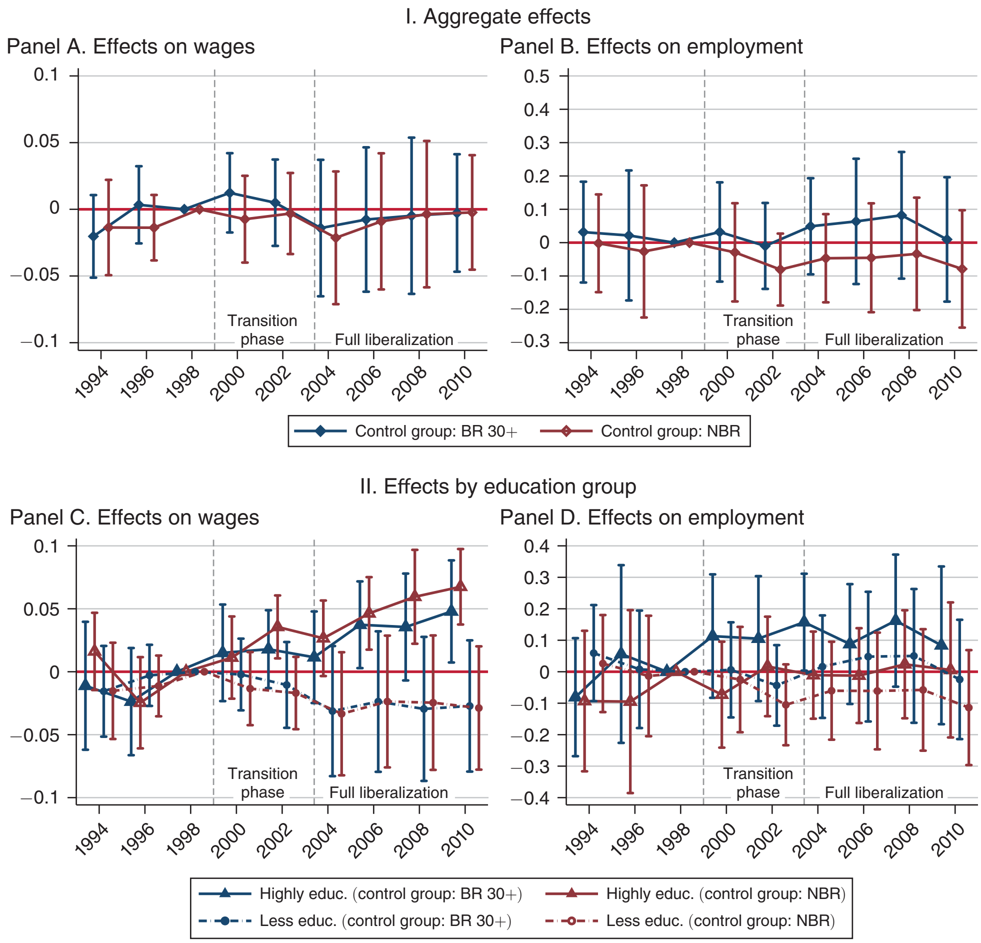

Figure 4. Effects of the Free Movement Policy on Wages and Employment of Natives

Notes: The figure shows the effects of the free movement policy on real wages and employment of native workers. It plots the event study coefficients and associated 95 percent confidence intervals for the highly treated municipalities based on equation (2). The dependent variables are natives' average log hourly real wage (panels A and C) and log total workers (panels B and D) per education group. We present separate estimates for the two control groups (BR 30+ and NBR). The regressions are weighted using the number of natives in a cell. Control variables are municipality and year fixed effects and linear NUTS-II trends. Standard errors are clustered by commuting zone.

This is not what we find. Rather, panel C of Figure 4 suggests a *positive* effect on real wages of highly educated natives that starts in the transition phase and grows to +4.5 percent in the free movement period. This positive wage effect is evident using either control group (cf. panels A and B of online Appendix Table A.5), it is robust, and economically significant: real wages of highly educated natives grew by only 3 percent, on average, in the BR between 1998 and 2010. The effects on employment for highly educated workers are imprecisely estimated but rule out strong negative employment effects. If anything, they suggest an *increase* in employment of highly educated native workers. On the other side, we find negative point estimates on some of the outcomes of less-educated natives. As standard errors are often quite large as well, however, none of the estimates is consistently statistically significant, and we cannot rule out zero effects on this group. Nevertheless, the heterogeneous point estimates by education group explain why we find a zero effect in the aggregate that averages those effects.

Overall, we find that highly educated natives gained from the increased availability of mostly highly educated CBW. This evidence is difficult to explain in a competitive labor market framework where immigration represents an increase in labor supply with a fixed labor demand. We develop this point below.

B. *Robustness and Extensions*

First, however, we discuss a series of important robustness checks for these labor market results. As in every DiD estimation, it is a first-order concern that the effects may be caused by a failure of common trends in outcomes across regions or by unobserved confounders. Potential confounders are changes in regional policies, unobserved demand, and productivity shocks that have a regionally unequal impact, e.g., due to differences in the industrial composition of regions. We address these concerns in several ways. In panel B of online Appendix Table A.6, we show that the estimated labor market effects are very close to the baseline estimates for the different outcomes if we include a Bartik (1991) variable, which controls for region-specific, sector-driven demand trends or shocks.[23] Similarly, the results are similar when we control for unobserved region-specific shocks using NUTS-II regions times year fixed effects (panel C of Table A.6), if we exclude the linear time trends contained in our baseline regression model (panel D of Table A.6), and if we drop the most important Swiss cities one by one (online Appendix Table A.7).[24] In Section VIE, we also present several pieces of evidence suggesting that our results are indeed attributable to increased labor mobility rather than caused by one of the other bilateral agreements that were signed at the same time.

Another concern with our labor market results is that they are driven by effects on the composition of native workers rather than by effects on "incumbent" workers in the regions of interest. The main concern in our case differs from the concern about native outflows in the existing literature, i.e., the fact that native workers may respond to immigrant inflows by leaving the labor market or the region (Borjas 2006; Dustmann, Schönberg, and Stuhler 2017). Our main result, an increase in wages of highly educated native workers while their employment is stable, could only be rationalized with *inflows* of high-wage natives from the control to the treatment regions. Unfortunately, we cannot examine flows of workers with the SESS because it does not follow individual workers over time. We thus investigate the question using data from the Swiss Labor Force Surveys (SLFS) conducted by the Swiss Federal Statistical Office. Overall, the analysis, presented in online Appendix Section C, provides no evidence that the greater availability of CBW affected the flow of natives between treated and control regions. In particular, we find no evidence of an increase in job-to-job transitions from the control to the treatment region,

[23] The basic intuition is to control for regional changes in employment or wages (by skill group), that are due to national-level changes in industries that are strongly represented in a particular region. See online Appendix Section B.3 on the construction of this variable.

[24] Omitting the linear time trends reduces the point estimate of the effect on wages highly educated natives and its precision. This point estimate, however, is not statistically different from the baseline estimate.

both for all workers and highly educated workers. Consistently, we find no influence of the reform on regional population size (column 6 of online Appendix Table A.9).

Beerli and Peri (2018) provide several further robustness checks. Most importantly, the paper shows that the results are similar if we only compare changes in outcomes in municipalities in the highly affected area (0–15 min) with outcomes in matched control municipalities that are similar in terms of predetermined characteristics (an approach also followed by Dustmann, Schönberg, and Stuhler 2017). The labor market results are also similar if skills are measured using occupation (high-, middle-, and low-paying occupations) instead of educational groups.

A useful extension to explain part of the identified wage effects is the important skill complementarity of natives and immigrants across occupations: highly educated CBW were primarily employed in technical, scientific, and engineering jobs while native workers had an incentive to move toward the higher end of managerial jobs (similar as in Peri and Sparber 2011; Peri, Shih, and Sparber 2015b). As management positions require knowledge of local culture, laws and norms, and possibly a local network of contacts, these positions are typically more accessible to natives than to foreigners.

We use the SLFS 1996–2009 to analyze whether the reform affected natives' probability to have a management position.[25] The SLFS identifies workers that are in the top executive level (Direktion/Geschäftsleitung) of a firm. The estimates in online Appendix Table A.10 suggest that the free movement policy increased the share and number of highly educated native workers in these positions by 7.2 percentage points: an 18 percent increase relative to the pre-reform average. We do not find evidence of an effect on the share of less-educated natives working in top management positions. Paralleling findings from Basten and Siegenthaler (2019) based on a different identification strategy, these results suggest that high-qualified native workers were more likely to become top-tier managers, possibly as a consequence of the imperfect substitutability with CBW.

How much of the positive reform effect on wages of high-qualified natives can be attributed to the increase in the share of them in better-paying management positions? We analyze this question in Table 4 using the SESS, which also contains information about the management rank associated to individuals' jobs. The first column reproduces our baseline effect on the wage of highly educated native workers in the free movement period. The next two columns show that this baseline effect is concentrated in higher management positions (column 2) in contrast to low/no management positions (column 3), corroborating the evidence that the reform increased the demand for workers in executive positions. In column 4, we calculate wages for highly educated natives holding the share of workers in management amounts to 70 percent of the actual wage effect (cf. columns 1 and 4). This simple calculation indicates that roughly 30 percent of the wage growth of high-qualified natives arises because they moved to higher management positions.

[25] We impose similar sample restrictions and use similar definitions regarding skill groups, worker characteristics, and geography as in the labor market analysis with the SESS data (see online Appendix Section B.1).

Table 4—Effect of the Free Movement Policy on Wages of Highly Educated Natives in Different Management Ranks

	All highly educated	Wage by management rank		Constant management rank shares
		High and middle	Low and no	
	(1)	(2)	(3)	(4)
$Free_t \times I(15 \leq d_i)$	0.045	0.054	0.014	0.033
	(0.015)	(0.016)	(0.016)	(0.011)
$Free_t \times I(15 < d_i \leq 30)$	0.015	0.025	−0.012	0.007
	(0.012)	(0.014)	(0.014)	(0.014)
Year and area fixed effects	✓	✓	✓	✓
NUTS-II trends	✓	✓	✓	✓

Notes: This table shows the effect of the free movement policy on mean log hourly real wages of highly educated natives in different management levels based on equation (1). Municipalities in the BR 30+ minutes constitute the control group (online Appendix Table A.11 provides the results with municipalities in the NBR as control group). Column 1 reports the baseline effect on all highly educated natives. In columns 2 and 3, highly educated natives are split into those with a high/middle and low/no management rank. Column 4 reports the effect on all highly educated when the share of high/middle managers is held at its 1998 level. This variable is the weighted average of the wages in high/middle positions p, $w_{m,t}^{p=h}$ and wages in low/no management positions, $w_{m,t}^{p=l}$ using the share of these groups' employment in 1998, $\gamma_{m',98} = L^{p=h}/L$ and $(1-\gamma_{m',98})$, as weights, i.e., $\tilde{w}_{m,t} = w_{m,t}^{p=h}\gamma_{m',98} + w_{m,t}^{p=l}$ $(1-\gamma_{m',98})$. $Free_t$ is 1 from year 2004 onward. $(d_i \leq x)$ and $(y < d_i \leq z)$ indicate whether a municipality is located less than x travel minutes or between y and z travel minutes from the next border crossing, respectively. Distance interactions with the transition phase are omitted for brevity. The regressions are weighted using the total number of natives in a cell. Robust standard errors, clustered by commuting zone, are given in parentheses.

V. Theoretical Framework

Our empirical findings in the previous sections show that opening the labor market for CBW in the municipalities close to the border produced an increase in the employment of these workers, especially the highly educated ones. Nevertheless, the reform led to an increase in the wages of highly educated native workers. These findings run counter to an interpretation of the reform as an increase in the local supply of high-skilled workers in a classic supply and demand framework within a perfectly competitive labor market (as used in Borjas 2014). In such a framework, an increase in high skilled foreign workers would decrease wages for high skilled native workers, at least in the short run. Moreover, such a framework is not well suited to describe how the removal of mobility restrictions affects job creation by facilitating firms' access to a pool of highly skilled and initially scarce workers.

We thus prefer to conceptualize our findings in an imperfect labor market framework with firm-employee matching, frictional search, and job posting (e.g., Pissarides 2000, Chassamboulli and Palivos 2014, Chassamboulli and Peri 2018). In such a model, firms post vacancies while workers with different skills search for jobs. The presence of search frictions leads to unemployment in equilibrium, and the labor market for each skill type is characterized by a specific ratio of vacancies to unemployment, called labor market tightness. The tighter the labor market for a particular skill type, the harder it is for firms to fill their vacancies. It is natural to assume that the labor market for certain specialized workers such as R&D workers is tighter.[26]

[26] As skilled workers have higher productivity and thus generate a higher surplus for the firm, firms will create a large number of job postings for them. At the same time, unemployment rates for skilled workers are lower due to long tenure in jobs (low break-up rates) and high specificity of matching.

Consequently, firms that depend more heavily on these workers will on average face a tighter labor market. These firms will experience "skill shortage."

In such a framework, the reform reduces the tightness of the labor markets for skilled workers by gradually increasing the availability of skilled CBW, which start to search for jobs in the Swiss labor market close to the border. The reform thus leads to a higher matching and vacancy-filling rate for skilled jobs with CBW. This, in turn, leads to an increase in employment of those workers and a decrease in unfilled vacancies of the firm. Because firms differ in the use of specialized workers, we expect that these effects are particularly pronounced in firms that were more dependent on workers that were scarce before the reform. Over time, firms will expand in response to the greater availability of CBW, and the number of vacancies will increase.

Such a model, however, cannot explain the observed *increase* in wages of high-skilled native workers: if the production function of the firm has constant or decreasing returns to scale, as in the standard Pissarides (2000) model, then the increase in employment of skilled CBW and the decrease in unfilled vacancies in the short run would *lower* wages of native skilled workers both through marginal productivity effects and through reducing skill-specific labor market tightness, which worsens workers' outside options when bargaining over wages. The observed positive wage effect on skilled natives, instead, requires that the firm's production function exhibits increasing returns to high-skilled workers. This is true if highly educated CBW have positive effects on firms' total or skill-specific productivity and/or physical capital. In this case, a greater availability of skilled workers can increase firms' surplus of a match, stimulate job creation, and thus lead to higher wages for high-skilled workers.

The literature suggests several channels through which a greater availability of high-skilled CBW may positively affect firms' productivity and capital formation. First, there may be static and dynamic productivity effects from increasing the number of skilled workers in a firm due to labor pooling, knowledge spillovers, and diffusion of ideas.[27] Similarly, an increase in R&D and other specialized workers could stimulate productivity because it is a direct input in the innovation process of firms and in the creation of new knowledge, either through increased patenting (Kerr 2013; Kerr, Kerr, and Lincoln 2014) or the exchange of ideas (see Hunt 2011, Hunt and Gauthier-Loiselle 2010). Second, an increase in specialized workers may trigger firms to adopt technologies that are better suited for using the skills supplied by CBW, increasing the specific productivity of that group (e.g., Peri 2012, Hornung 2014, Lewis 2011). The evidence, presented in Section IVB, that skilled CBW and natives specialized in different jobs can also attenuate possible negative effects on the marginal productivity of skilled native workers and enhance the positive wage effects on them from productivity changes. Third, an increase in high skilled workers may attract physical capital and induce firm-creation and investment (Kerr, Kerr, and Lincoln 2015; Olney 2013).[28] If capital and skilled workers are complements

[27] Moretti (2004) and Diamond (2016), among others, show that a larger share of college-educated workers increases labor productivity in US cities. Glaeser and Mare (2001) show that this may be in part due to dynamic local learning. Iranzo and Peri (2009) argue that it may be the consequence of faster technology adoption.

[28] Theoretically, we expect that firms' entry and location choices under nonzero profits depend upon the same quantities as those that affect firms' sales and profits (Combes and Gobillon 2015). The prospect of hiring the right type of workers can be a strong attractor for firms and a key driver of agglomeration economies (as in Moretti 2004).

(Krusell et al. 2000), the resulting increase in physical capital in the area would contribute to increased wages of skilled native workers.

Overall, the combination of search frictions in the (skilled) labor market, heterogeneous firm-level dependence on these workers, and positive productivity effects of high-skilled CBW are the key ingredients to understand our empirical findings. These theoretical considerations lead to a series of predictions regarding the effect of the policy that we test in the following section using firm-level data.

VI. Mechanisms: Firm-Level Analysis

A couple of remarks on the firm-level regressions are warranted before turning to these analyses.[29] First, the unit of observation in the regressions is an establishment (in the Business Census, BC) or a firm (in the Innovation Surveys, IS). Second, we focus on incumbent firms because the free movement policy changed the composition of firms (as shown in Section VID). In the BC, we thus restrict the sample to a fully balanced panel by focusing on establishments that existed in all sample periods. In the IS, balancing the panel would lead to a very small sample ($n \approx 100$), so we focus on incumbents by restricting the sample to firms that existed before 1999. As can be seen in online Appendix Table A.16, alternatives to account for firm entry lead to very similar results.[30] Third, to ensure that the sample is comparable to what we used in the labor market analysis, we focus on private-sector establishments and drop those with less than three FTE workers in the first census. For the same reason, we weight regressions based on the BC by establishment's average size over the sample. In the IS, however, our preferred regressions are not weighted by firm size.[31] Fourth, in both datasets, we drop a very small number of extreme outliers, which significantly affect the precision, but not the estimate of the coefficient of interest (as shown in online Appendix Table A.16). Finally, for brevity, the control group will mainly be firms in the BR 30+ minutes away from the border. Both control groups lead to similar results with few exceptions that we discuss (online Appendix Figure A.1 and Table A.16 provide a direct comparison).

A. *Effects by Skill Intensity of Industries*

This section assesses a first set of predictions from our theoretical considerations: the greater availability of workers should lead to additional hiring of CBW and stimulate the creation of additional jobs within firms. Since most new CBW are skilled, these effects should be most prevalent in firms whose production is more intensive

[29] Online Appendix Section B.2 contains a detailed discussion on the construction of our firm-/establishment-level estimation samples. For ease of exposition and brevity, we mainly focus on the preferred specifications of Ruffner and Siegenthaler (2017). We refer the reader to this working paper for a full discussion on how the specification choices influence the main results.

[30] One is to include *all* firms that fulfill the other sample restrictions (i.e., to allow firm entry and exit) and to account for entry using the firm/establishment fixed effects only (panel G). Another is to follow a single cross-section of firms (such as those existing in 1998, panel H).

[31] Weighting by firm size is unattractive in the IS because it gives a large weight to the few, large multi-establishment firms in the survey. In the IS, we assign multi-establishment firms to the location of the headquarter, which leads to mismeasurement (see Section IIA). Online Appendix Table A.18 compares the weighted and unweighted regression results. The effects have the same sign, but the former are larger and more sensitive to the choice of specification.

in skilled workers. To test these predictions, we examine how the reform affected employment in establishments separately for low- and high-tech manufacturers, and knowledge-intensive (finance, business, human resource management) and more traditional service sector establishments.

The results, based on the balanced panel of establishments from the BC, are presented in Figure 5. The dependent variable in panel A is log FTE employment (panel A). In panels B and C, the outcome is FTE employment of foreign workers (i.e., resident immigrants plus CBW) and Swiss workers, respectively. For consistency with the results in Section III, we express these outcomes as shares of total FTE employment in 1998, but the results are comparable if we use an approximate log outcome (see online Appendix Figure A.3). Because the BC in 1991 and 2011 do not contain information on workers' nationality, panels B and C are restricted to the census waves 1995–2008.

Figure 5, first, suggests that highly treated establishments and control establishments displayed a similar within-establishment change in all these outcomes in the pre-reform period. According to panel B, high-treatment incumbent establishments began to hire more foreign workers than the control group starting in the 1998–2001 period. Consistent with our theoretical considerations, the hiring mainly happens in high-tech manufacturing and the knowledge-intensive service sector. As shown by panel C, the increased employment of foreigners in skill-intensive sectors did not lead to a reduction in employment of Swiss nationals in the firms. Instead, highly treated incumbent establishments seem to have created *additional* jobs for the newly hired foreigners. This happened as early as in the transition phase, simultaneously to the increase in employment of CBW. By 2011, highly treated establishments in the two skill-intensive sectors are substantially larger than control establishments in the same sector.

Overall, we find strong evidence that the greater availability of CBW led to the creation of additional jobs for the newly available foreign workers. Averaged across all sectors, the free movement policy increased foreign employment as a share of 1998 employment in highly treated establishments by 8.5 percent and total FTE employment by 6.2 percent, as shown in columns 1 and 2 of Table 5. The estimated reform effect on the size of incumbent firms in the IS is even slightly larger (column 3 of Table 5). Moreover, we generally find similar results if we use firms in the NBR as control group except that the effects on establishment size in the BC are almost exclusively driven by high-tech manufacturers (see online Appendix Figures A.1 and A.2).

Table 6 tests another implication of our theoretical considerations: if the reform's positive wage effects are due to increased productivity or capital accumulation in firms, the sectors that have the largest increase in employment of CBW should also account for natives' wage gains.

The table shows that the wage increase of natives are indeed concentrated in high-tech manufacturing and the knowledge-intensive service sector. Interestingly, if we estimate the sector-specific effects separately by natives' educational attainment, we find that the wage gains in the knowledge-intensive service sector accrue to highly educated natives only. The evidence is more mixed in high-tech manufacturing along this dimension: the wage gains accrue to less-educated natives in the highly treated region but to highly educated natives in the *weakly* treated

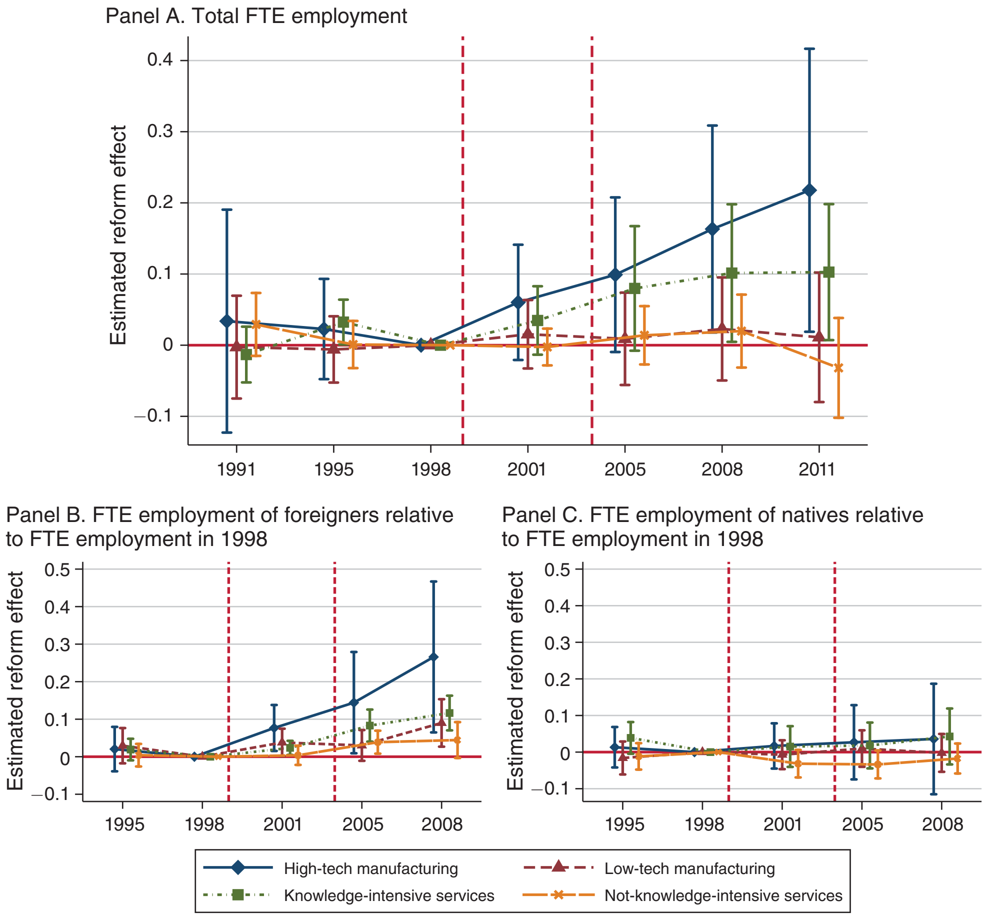

Figure 5. Effects of the Free Movement Policy on FTE Employment of Incumbent Establishments

Notes: The figure shows the effect of the free movement policy on FTE employment of incumbent establishments using private-sector establishment-level data from the BC. It shows event study coefficients and associated 95 percent confidence intervals for highly treated establishments based on equation (2), estimated separately by establishments' broad sector of economic activity. The regressions control for establishment fixed effects, year fixed effects, and linear NUTS-II trends. The control group is establishments in the BR with more than 30 minutes travel distance to the border (the results using the other control group are presented in online Appendix Figure A.2). In panel A, the dependent variable is log FTE employment. In panel B, it is FTE employment of foreigners as a share of total employment in 1998. In panel C, it is FTE employment of Swiss nationals as a share of total employment in 1998. The samples in panels B and C cover the 1995–2008 period because the BC in 1991 and 2011 do not contain information on workers' nationality. All regressions are weighted using average establishment size (in FTE). Standard errors are clustered by commuting zone.

region.[32] In contrast, we consistently find no wage effects on natives in the two sectors that are not skill-intensive. In fact, Table 6 provides evidence for non-negligible wage *declines* among the less-educated natives in non-knowledge intensive service

[32]The results are similar if we use municipalities in the NBR as control group (see online Appendix Table A.12). Why may less-educated natives have benefited from the reform in high-tech manufacturing but not in knowledge-intensive services? Differences in immigrant inflows by skills cannot explain this pattern: online Appendix Table A.13 shows that the skill ratios of immigrants are roughly similar in both sectors. One plausible answer is thus differences in complementarities in production across sectors.

Table 5—Effects of the Free Movement Policy on Various Firm Outcomes

	Foreign employment	Establishment size (FTE)	Firm size (FTE)	Sales	Productivity	Patent applications 0/1	Patent applications IHS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Transition_t \times I(d_i \leq 15)$	0.025	0.024	0.015	−0.004	−0.001	0.017	0.038
	(0.009)	(0.017)	(0.033)	(0.036)	(0.036)	(0.019)	(0.028)
$Transition_t \times I(15 < d_i \leq 30)$	0.031	0.049	0.035	−0.008	−0.044	0.004	−0.022
	(0.007)	(0.016)	(0.029)	(0.033)	(0.035)	(0.015)	(0.028)
$Free_t \times I(d_i \leq 15)$	0.085	0.062	0.098	0.120	0.037	0.064	0.126
	(0.020)	(0.022)	(0.046)	(0.050)	(0.035)	(0.027)	(0.046)
$Free_t \times I(15 < d_i \leq 30)$	0.034	0.055	0.091	0.049	−0.042	0.018	0.039
	(0.012)	(0.023)	(0.046)	(0.044)	(0.039)	(0.024)	(0.050)
Observations	252,962	358,619	9,250	8,456	7,107	9,032	8,901
R^2	0.616	0.955	0.965	0.973	0.729	0.710	0.820
Dataset	BC	BC	IS	IS	IS	IS	IS
Sample period	95–08	91–11	95–12	95–12	95–12	96–13	96–13
Period effects	✓	✓	✓	✓	✓	✓	✓
Firm/establishment effects	✓	✓	✓	✓	✓	✓	✓
NUTS-II trends	✓	✓	✓	✓	✓	✓	✓
Weights	✓	✓					
Number of clusters	72	72	73	73	71	73	73

Notes: The table presents results of establishment- and firm-level DiD regressions using the BC (columns 1 and 2) and the IS (columns 3–7). The control group are firms in the BR with more than 30 minutes to the border (results using the other control group are presented in panel B of online Appendix Table A.16). The dependent variable in column 1 is full-time equivalent (FTE) employment of foreigners as a share of total employment in 1998. The dependent variable in column 2 is log FTE employment. The dependent variables in columns 3–5 are firms' log FTE employment, log total sales, and log value added per FTE worker. The dependent variable in column 6 is a dummy equal to 1 if a firm filed at least one patent application in the year of the survey and the two years before the survey. Column 7 uses the inverse hyperbolic sine (IHS) of the number of patent applications. $Transition_t$ is a dummy equal to 1 between 1999 and 2003, whereas $Free_t$ is 1 from year 2004 onward. $I(d_i \leq x)$ and $I(y < d_i \leq z)$ indicate whether a firm is located less than x travel minutes or between y and z travel minutes from the next border crossing, respectively. All regressions account for establishment (BC) or firm (IS) fixed effects, period fixed effects, and linear trends per NUTS-II region. The regressions in columns 1 and 2 are weighted using average establishment size (in FTE) as weight. Standard errors are clustered by commuting zone.

industries. Those sectors are potentially those where capital and productivity effects were less pronounced, as technology is more traditional. Therefore, the increased labor market competition, especially from less-educated CBW, might have prevailed in the short run.

B. *Firm Productivity*

Can we explain the wage increases of natives in the skill-intensive sectors established in the previous section with productivity gains in incumbent firms in these sectors, as hypothesized in Section V? As a first step to answer this question, columns 4 and 5 of Table 5 present the results of firm-level regressions of equation (1) using the log of firms' sales and value added per FTE worker as dependent variables.[33] The data stem from the IS 1996–2013 but refer to the year before the survey. Column 4 suggests that the free movement policy increased sales of highly treated incumbent firms by almost 12 percent. Despite this sizable positive effect

[33] The associated event study results are shown in online Appendix Figure A.1.

Table 6—Effects of the Free Movement Policy on Wage Levels of Natives by Sector of Employment

	Manufacturing		Services	
	High-tech (1)	Low-tech (2)	Knowledge-intensive (3)	Not-knowledge-intensive (4)
Panel A. All education groups				
$Free_t \times I(d_i \leq 15)$	0.050	0.015	0.040	−0.028
	(0.016)	(0.018)	(0.017)	(0.022)
$Free_t \times I(15 < d_i \leq 30)$	0.016	0.014	0.024	−0.018
	(0.017)	(0.016)	(0.012)	(0.014)
Panel B. Highly educated				
$Free_t \times I(d_i \leq 15)$	0.001	−0.032	0.083	0.031
	(0.015)	(0.034)	(0.017)	(0.020)
$Free_t \times I(15 < d_i \leq 30)$	0.030	−0.011	0.029	−0.014
	(0.014)	(0.024)	(0.016)	(0.029)
Panel C. Less-educated				
$Free_t \times I(d_i \leq 15)$	0.042	0.020	0.009	−0.037
	(0.014)	(0.018)	(0.017)	(0.020)
$Free_t \times I(15 < d_i \leq 30)$	0.005	0.016	0.008	−0.029
	(0.017)	(0.017)	(0.012)	(0.014)
Year and area fixed effects	✓	✓	✓	✓
NUTS-II trends	✓	✓	✓	✓

Notes: This table shows the effects of the free movement policy on mean log hourly real wages of natives by education group and sector of employment based on equation (1). Municipalities in the BR 30+ constitute the control group (results using the other control group are presented in online Appendix Table A.12). High-tech manufacturing is NACE Rev 1.1 industries 24, 29, 30, 31, 32, 33, 34, and 35 excluding 35.1. Low-tech manufacturers are the other manufacturing industries. Knowledge-intensive services are NACE Rev 1.1 industries 61, 62, 64, 65–67, 70–74, 80, 85, 92. Not knowledge-intensive services are the rest of the service sector industries. $Free_t$ is 1 from year 2004 onward. $(d_i \leq x)$ and $(y < d_i \leq z)$ indicate whether a municipality is located less than x travel minutes or between y and z travel minutes from the next border crossing, respectively. Distance interactions with the transition phase are omitted for brevity. The regressions are weighted using the total number of natives in a cell. Robust standard errors, clustered by commuting zone, are shown in parentheses.

on sales, the liberalizations did not have a statistically significant impact on labor productivity of the average firm (column 5).

However, this average effect masks an increase in productivity in the skill-intensive sectors that accounts for the positive wage effect on natives. Panel A of Figure 6 presents the results of an augmented version of our baseline regression (column 5 of Table 5) that contains interactions between the treatment variables and dummy variables equal to 1 if a firm belongs to the two skill-intensive sectors. The regression controls for detailed industry-period fixed effects so that we only compare firms in the same industry across regions. The figure suggests that the reform had a substantial positive effect on labor productivity of incumbent high-tech manufacturers in the free movement period. We also find a positive reform effect on the knowledge-intensive business service firms, but it is only statistically significant at the 10 percent level.

Another central prediction of our theoretical framework is that the productivity effects of the reform should be more pronounced in firms that faced particularly tight labor markets for skilled workers before the reform. We test this prediction leveraging the fact that firms in the IS were asked directly whether their innovation efforts were negatively affected by a "shortage of specialized personnel." We average the

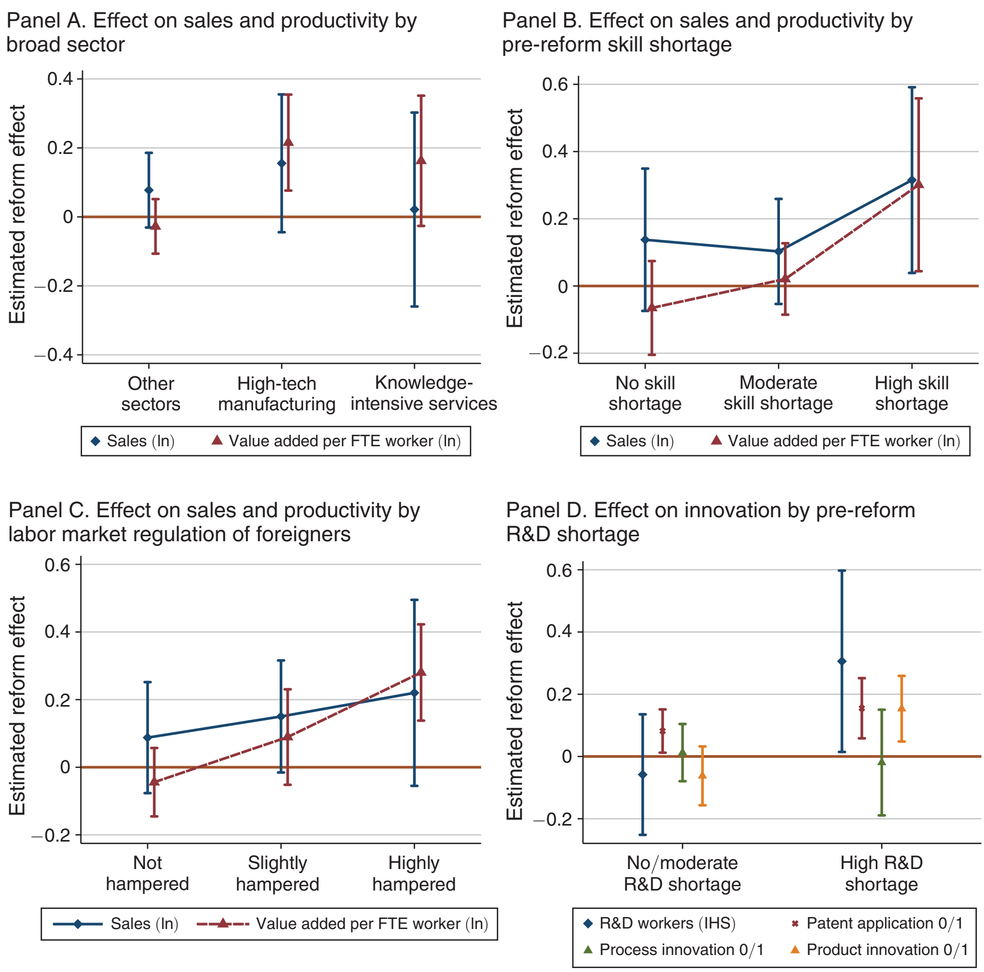

Figure 6. Effects of the Free Movement Policy on Firms' Sales, Productivity, and Innovation Outcomes by Dependence on Skilled Workers

Notes: The figure examines whether the effects of the free movement policy on firms' sales, productivity, and innovation depend on the skill intensity of the sector and firms' perceptions to be constrained by labor shortages prior to the reform. All regressions are based on firm-level data from the IS 1996–2013, focus on the effects on highly treated firms in the free movement phase, and use firms in the BR located more than 30 minutes to the border as control group. The coefficients are derived from our baseline regression model (equation (1)) augmented with interactions between all reform variables and certain indicators. In panel A, the indicators refer to knowledge-intensive service industries and high-tech manufacturers. In panels B, C, and D, the indicators refer to firms that differed in the extent to which their innovation efforts were negatively affected by a shortage of specialized personnel (panel B), labor market regulation for foreigners (panel C), and shortage of R&D personnel (panel D) in the IS 1996 and/or 1999. The dependent variables in panels A–C are log total sales and log value added per FTE worker. The dependent variables in panel D are the Inverse Hyperbolic Sine (IHS) of the number of R&D workers, and dummies equal to 1 if a firm filed at least one patent application, reports to have introduced at least one process, or at least one product innovation in the three years preceding the survey. All regressions control for firm fixed effects, year fixed effects, and NUTS-II trends. Panel A also controls for industry-period effects. The vertical bars represent 95 percent confidence intervals constructed based on standard errors clustered by commuting zone.

original, five-point Likert scale survey item over the two survey waves prior to the reform for each firm and group firms into three categories, from "no shortage" to "high shortage."[34] As shown in Table 2, 17 percent of all highly treated firms and similar shares in the control regions experienced high skill shortages before the reform. Interestingly, skill shortages appear to be broadly distributed across different segments of the economy: no single firm characteristic correlates strongly with firms' pre-reform problems to find skilled workers.[35] If we interact the reform variables with the indicators of skill shortages, as is done in panel B of Figure 6 to test our theoretical prediction, we find strong evidence that the reform led to productivity gains in highly treated firms with skill shortages prior to the reform.

Panel C of Figure 6 leverages a similar subjective survey question to differentiate between firms that differed in the extent to which they reported that their innovation activities were hampered by "labor market regulation for foreigners" prior to the policy changes. Panel C of Figure 6 suggests that relaxing this obstacle spurred productivity growth in firms that were constrained initially.

Overall, the findings presented in this section support the view that the wage gains of natives were the consequence of increased productivity in firms with a high demand for skilled workers. They are also consistent with results from studies on the H-1B program in the United States, which typically find that changes in the number of H-1B visas nationally affect productivity in firms and regions that rely strongly on H-1B workers (Ghosh, Mayda, and Ortega 2014; Kerr and Lincoln 2010; Peri, Shih, and Sparber 2015a).

C. *Innovation*

As discussed in Section V, another possible explanation for the positive wage effects of the open border policy is that the highly skilled CBW supported firms' innovation activities. To study this channel, we first analyze whether the newly hired CBW played an important role in the growth of research and development (R&D) departments of firms close to the border. Using data from the SESS, panel A of Figure 7 plots the share of researchers in total employment depending on the travel distance to the border in 1996, 2000, and 2010. It illustrates that cross-border researchers represented almost one-third of all R&D workers in firms located within 15 minutes to the border in 2010. As for CBW in general, the presence of cross-border researchers declines strongly with distance to the border. The figure also documents an increase in the share of cross-border researchers between 2000 and 2010 that is concentrated close to the border, suggesting that the free movement policy increased the total R&D employment share.

The regression analysis supports a causal interpretation of this effect. At the municipal level, we find that the free movement policy increased the share of immigrant researchers in total employment by about 0.6 percentage points (see online Appendix Table A.4). We study the effect of the policy on total R&D employment at the firm level in panel D of Figure 6 using data from the IS 1996–2013. The dependent variable is the inverse hyperbolic sine (IHS) of the number of R&D

[34] In particular, firms that have "no shortage" are firms with a less than 2, "moderate shortage" firms have a value between 2 and 4, and "high shortage" firms have a value greater than or equal to 4.

[35] See online Appendix Table A.14. Firms that reported moderate or high shortages are slightly more likely to be manufacturers and to have R&D expenditures.

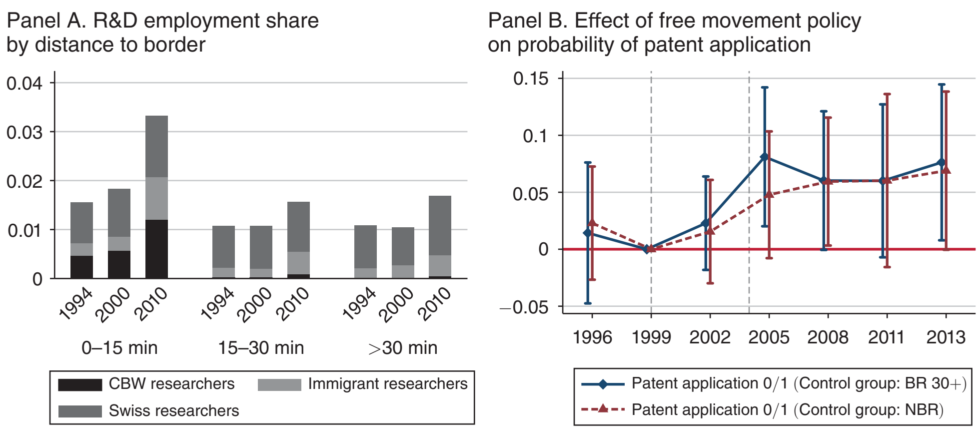

FIGURE 7. EFFECT OF THE FREE MOVEMENT POLICY ON R&D EMPLOYMENT AND PATENTING

Notes: Panel A plots the employment share of workers in the occupation "research and development" by immigrant status and travel duration to the nearest border crossing using data from the SESS in 1994, 2000, and 2010. The figure focuses on private-sector firms in the BR. Panel B plots the sequence of effects, and associated 95 percent confidence intervals, of the free movement policy on highly treated firms using firm-level data from the IS 1996–2013. The two event studies are based on separate regressions of equation (2) with the two control groups (municipalities in the BR located more than 30 travel minutes to the border and municipalities in the NBR). The dependent variable is a dummy equal to 1 if a firm filed at least one patent application in the year of the survey or the two years before the survey. The regressions control for firm fixed effects, year fixed effects, and NUTS-II trends. Standard errors are clustered by commuting zone.

workers.[36] The estimates thus reflect an approximate percentage increase. Guided by our theoretical considerations, we estimate separate effects for firms that were more or less affected by the tightness of the labor market for R&D workers prior to the reform. We distinguish firms that differed in the extent to which their innovation efforts were "hampered by lack of R&D workers" in the IS 1996 and/or 1999.[37] The latter represent 1/6 of all firms (see Table 2). The results in panel D of Figure 6 suggest that the policy increased R&D employment in incumbent firms. As expected, the effect is driven by firms with more severe shortages of R&D workers before the reform.

Did the increase in R&D employment translate into more inventions? The event study depicted in panel B of Figure 7 examines whether the reform affected the probability that highly treated firms filed at least one patent application in the three years before the IS. The figure provides robust evidence that the reform increased this probability. The estimated reform effect is approximately 6 percentage points relative to both control groups. The effect on the IHS of the *count* of patent applications is around 12 percent (see column 7 of Table 5).[38] The difference in patent applications between firms in the highly treated and the control regions arises

[36] The inverse hyperbolic sine (IHS) of the number of patents accounts for the substantial amount of firms without patents and the long right tail of the distribution. The IHS of outcome y is $IHS(y) = \ln\left(y + \sqrt{1 + y^2}\right)$. As argued by Doran, Gelber, and Isen (2015), using the IHS is attractive for innovation outcomes because it approximates the log of an outcome but has the advantage that it is defined at 0.

[37] Empirically, the effects for firms with no or moderate pre-reform R&D shortage differ little, so we pool the two groups in the graph.

[38] The corresponding event study is presented in panel D of online Appendix Figure A.1. Online Appendix Table A.15 shows heterogeneity analyses.

mainly between 2002 and 2005. Prior to the reform and from 2005 onward, the trends in patent applications are comparable. As shown in panel D of Figure 6, the effect on patenting is larger for firms that reported shortages of R&D workers prior to the reform, consistent with the larger growth of R&D employment in these firms.

Finally, panel D of Figure 6 also shows that the reform affected the actual *output* of the innovation process. In particular, we find that the reform had a positive impact on the probability that firms report a product innovation in the IS. Product innovations are defined as the introduction of a good or service that is either new or a substantially improved version of a prior good or service. By contrast, we find no measurable impact on the probability to report process innovations, even among firms that lacked R&D workers before the reform.

Overall, greater access to CBW seems to have increased employment of R&D workers, patenting, and product innovations. These effects are concentrated among incumbent firms that declared shortages of R&D workers before the policy implementation. Additional results reported in Ruffner and Siegenthaler (2017) show that the increase in employment of cross-border researchers had no measurable effect on employment and wages of native researchers and may have *crowded in* other resident immigrant researchers. These results add to an unsettled debate on whether inflows of skilled immigrants benefit high-skilled residents. Different studies on the impacts of H-1B workers or foreign-born scientists reach conflicting conclusions.[39] In the private sector context analyzed here, firms seem to have reacted to the greater availability of R&D workers by creating additional R&D jobs, which helped to absorb the increased supply of CBW.

D. *Establishment Entry and Exit*

Our theoretical considerations suggest that the increased availability of CBW could lead to capital formation. If capital is complementary to high skilled workers, capital deepening could also have contributed to the growth of the wages of highly educated workers.

An indirect way of assessing the relevance of this channel is to test whether the free movement policy led to entry and reduced exits of establishments. We use the BC to study this question. An establishment is considered an entrant (an exiter) if its establishment identifier is new (disappears). The estimations are run at the municipal level and cover the period 1991–2008 in the case of firm entry and the period 1991–2011 in the case of firm exit.[40] Figure 8 presents separate event study regressions for the two control groups based on this information. The outcome

[39] The results in Ghosh, Mayda, and Ortega (2014) and Kerr and Lincoln (2010) suggest that greater access to H-1B workers generally increases the size, productivity, and innovation performance of firms that rely heavily on H-1B visas. Doran, Gelber, and Isen (2015), on the other hand, find that winning an additional H-1B worker in the H-1B lotteries of 2006 and 2007 increased firms' profits, had no effect on patenting and firm size and crowded out resident workers. Similarly, Kerr, Kerr, and Lincoln (2015) find that hiring young skilled immigrants increases firms' skill intensity, but their evidence regarding firm size is inconclusive. Finally, Borjas and Doran (2012) find that the strong influx of Russian mathematicians after the collapse of the Soviet Union had negative effects on the academic positions of US mathematicians. As pointed out by Card and Peri (2016), immigrant and resident mathematicians likely compete for a fixed number of positions in academia even in the medium run, which increases the scope for displacement.

[40] The reason for the difference in sample coverage is given in online Appendix Section B.2.

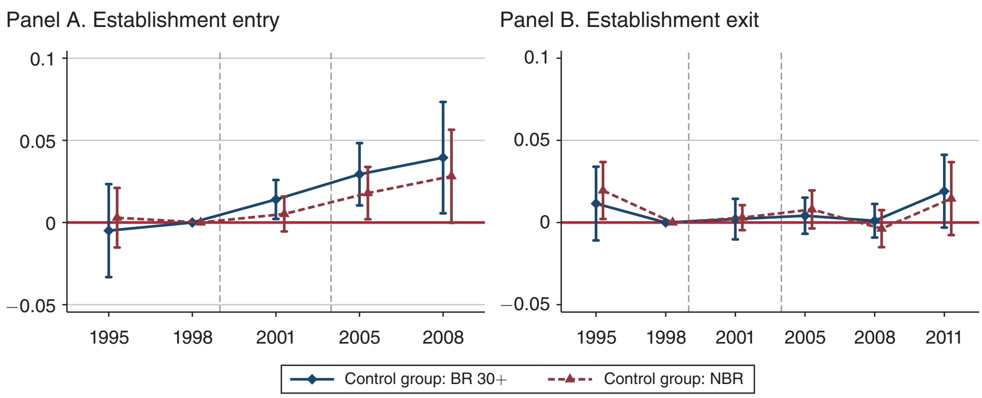

Figure 8. Effects of the Free Movement Policy on Establishment Entry and Exit

Notes: The figure plots the sequence of effects, and associated 95 percent confidence intervals, of the free movement policy on highly treated municipalities. The two event studies per panel are based on separate regressions of equation (2) with the two control groups (municipalities in the BR located more than 30 travel minutes to the border and municipalities in the NBR). In panel A, the dependent variable is the number of new establishments in t as a fraction of the number of establishments in 1998. The estimation sample is based on municipality-level data from the BC 1991–2008. In panel B, the dependent variable is the number of establishments exiting between $t-1$ and t as a fraction of the number of establishments in 1998. The estimation sample is based on the BC 1991–2011. The regressions are weighted using the number of establishments in 1998 per municipality as weight. All regressions include municipality and year fixed effects and NUTS-II trends. Standard errors are clustered by commuting zone.

variable in panel A is the number of establishments entering a municipality between BC waves $t-1$ and t relative to the number of establishments in 1998. The outcome variable in panel B is the number of establishment exits per municipality relative to the number of establishments in 1998. The regressions are weighted by the number of establishments in a municipality in 1998.

The figure provides evidence that the reform had a positive impact on establishment entry. By 2008, the policy seems to have increased the share of new establishments in the highly treated region by roughly 4 percentage points relative to 1998. Importantly, this firm entry appears to have occurred early during the reform. It starts in the 2001–2005 period if we use municipalities in the NBR as the control group, and even in the 1998–2001 period if we use firms in the BR located far away from the border. In line with our previous results, the increase in firm-creation was most pronounced in high-tech manufacturing and in knowledge-intensive services, as shown in columns 2–5 of online Appendix Table A.21. We find no evidence that the reform affected establishment exit (panel B of Figure 8). This finding also limits the danger that our firm-level regressions are affected by survivorship bias.[41]

Overall, the results on net firm entry indicate that the greater access to CBW encouraged capital formation.[42] These results are consistent with those of Olney

[41] This bias could mean that we attribute too much, or not enough, of the reform effect to occur within firms rather than to the change in firms' composition. Ruffner and Siegenthaler (2017) present additional robustness checks to address this concern, e.g., by constructing lower bounds on the reform effects using a trimming procedure proposed by Lee (2009).

[42] We also find indirect evidence for such an effect when studying the staffing decisions of multi-establishment firms. Online Appendix Table A.22 presents DiD estimates that focus only on multi-establishment firms with branches in different treatment regions. We find that the reform increased the growth of establishments located close to the border relative to establishments within the *same* firm farther away from the border. This suggests that

(2013), who finds a quick response of establishment formation in low-skill intensive industries to an immigration-induced shock in low-skill labor supply. They are inconsistent, however, with evidence suggesting that spatial adjustments to immigration may take a long time to occur (see Jaeger, Ruist, and Stuhler 2019).

E. *Robustness of Firm Results*

We perform a similar set of robustness checks for our firm results as for the labor market results. These checks reveal that our estimates are robust to the exclusion of the most important large cities (online Appendix Table A.19) and the exclusion of the linear time trends contained in our baseline regression model (panel C of online Appendix Table A.16). The estimated reform effects are also similar if we add industry-period, NUTS-II-period, and canton-period fixed effects (panels D–F of Table A.16), controls that absorb many unobserved region- and industry-specific shocks that could confound the main results.

We paid particular attention to ensure that the estimated effects are not driven by changes in trade and market access, and, more generally, by the other bilateral agreements signed at the same time as the AFMP.[43] As mentioned above, one of these agreements aimed at facilitating trade with the EU by reducing non-tariff barriers to trade. Table 2 shows that the share of exporters and importers is slightly larger in regions close to the border. If the agreement affected firm and productivity growth in Switzerland, the effects are plausibly larger in regions close to the border.

Several pieces of evidence suggest that neither trade flows nor any of the other bilateral agreements drive our results. First, we find very limited evidence that the reform effects on firms were larger for firms that export more (see Figure A.4 and panel C of Table A.15 in the online Appendix). Second, the reform had no effects on firms' export status and export intensity. Rather, the results suggest that exports grew in parallel to domestic sales of treated firms (online Appendix Table A.23). Third, our firm and labor market results are similar, if anything even stronger, if we exclude all two-digit industries that were directly affected by the other bilateral agreements (see panel D of online Appendix Table A.6 and panel F of Table A.16).[44] Fourth, using a set of qualitative questions in the IS about which policy-related factors hampered firms' innovation efforts, we show that there were no other changes in firms' political and regulatory environment (such as the access to the EU market) that changed differently between treated and control firms in the period of interest. Only one factor correlates systematically with the variables of interest (see online Appendix Table A.24): the probability that firms perceive their innovation activities hampered by "labor market regulation for foreigners." This is likely a direct consequence of the labor market liberalizations.

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multi-establishment firms responded to the policy by increasing the size of establishments where CBW became more easily available.

[43] In Ruffner and Siegenthaler (2017), we also check that the results are not driven by movements in the exchange rate. In general, the exchange rate remained relatively constant in the period of interest in this paper.

[44] We proxy exposure to these agreements using a classification created by Bühler, Helm, and Lechner (2011) who study how the trade liberalization caused by the bilateral agreements affected plant growth in Switzerland. The authors carefully assess the extent to which a specific two-digit industry was affected by the six other bilateral agreements next to the free movement agreement. The authors assign industries into three categories: not affected, affected, and strongly affected. In the table, we only keep non-affected industries.

Overall, these results suggest that factors such as the other bilateral agreements are unlikely to be a main driving force behind our results. Moreover, several of our heterogeneity analyses, such as the results in Figure 6, lend credibility to our empirical strategy: in the cross section of firms, the reform effects are generally largest among firms that we may expect to profit most from better access to skilled foreign workers. Nevertheless, we cannot rule out that there are some other factors that may have contributed to differences in the growth of certain outcomes between the regions of interest, especially given the relatively gradual implementation of the policy and our focus on longer-term outcomes.

VII. Conclusion

This study sheds light on the effects of fully opening the Swiss labor market to CBW on the number and types of cross-border workers hired in Switzerland, on wages and employment of native workers, and on the number, employment, sales, productivity, and innovation activities of Swiss firms. Empirically, we exploit the sequential introduction of the free movement of persons and the fact that CBW mobility affected Swiss regions close to the border earlier and more strongly.

We show that the greater availability of CBW produced a progressive and significant increase in their employment in municipalities close to the border, while it had virtually no effect beyond 30 minutes driving distance from the border. Nevertheless, natives working in municipalities close to the border did not experience a statistically significant reduction in average log hourly wages and log employment, relative to municipalities further away from the border. Instead, we find evidence that wages of highly educated natives *increased* as a consequence of the reform, despite the fact that two-thirds of the new CBW were also highly educated. We show that the positive wage effect resulted from pushing some of the natives to managerial and high-paying occupations, from stimulating productivity and job growth in incumbent firms, from attracting new firms, and from promoting firms' innovation activities. Most of these effects are most prominent in skill-intensive sectors and in firms that revealed shortages of skilled workers before the reform. Overall, our evidence is consistent with growth in the number of jobs sufficient to absorb the CBW inflow without native displacement or wage decline.

Our results have at least three important insights that future research could extend. First, they highlight the important role of firms in determining the labor market effects of immigration. The policy changes affected the dynamics of firm entry, firm growth, and productivity, indicating that firms recognized the opportunities created from better access to highly skilled CBW. Second, our findings corroborate claims of business leaders that opening the border for foreign workers can benefit firms' performance. There has been little systematic research whether and which firms profit from increased access to foreign workers. Third, our results suggest that the gradual and predictable implementation of the reform may have played a central role in enabling adjustments by firms that allowed absorbing the increased supply of CBW. We encourage further studies that focus on changes in immigration policies to gain insights on how immigration policy can foster firms' success without harming the labor market opportunities of native workers.

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